



**DEPARTMENT OF THE ARMY
UNITED STATES ARMY INFORMATION
SYSTEMS ENGINEERING COMMAND
FORT HUACHUCA, ARIZONA 85613-5300**



**SECRET INTERNET PROTOCOL ROUTER
NETWORK
TECHNICAL IMPLEMENTATION CRITERIA,
VERSION 7.0**

SEPTEMBER 2013

FORT DETRICK ENGINEERING DIRECTORATE

Distribution C

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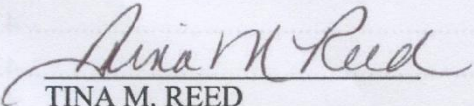
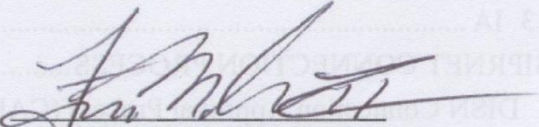
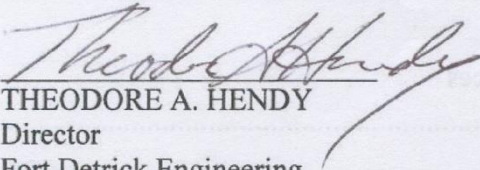
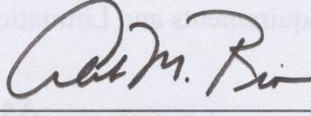
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TABLE OF CONTENTS

	Page
1.0 INTRODUCTION.....	1
1.1 Purpose.....	1
1.2 Scope.....	1
1.3 SIPRNET Project Goal	1
2.0 MAJOR PARTICIPANTS AND RESPONSIBILITIES	1
2.1 U.S. Army Corps of Engineer (USACE).....	1
2.2 U.S. Army Information Systems Engineering Command (USAISEC)	2
2.3 USACTTA.....	2
2.4 DAA	3
2.5 NEC/Communications Service Provider	3
2.6 User IMO	4
2.7 Shared	4
3.0 TECHNICAL SOLUTIONS DESCRIPTION AND CRITERIA	5
3.1 Construction Criteria.....	5
3.2 Connectivity Areas of Responsibility	6
3.3 Access Areas	10
3.4 PDS	13
3.5 SIPRNet Architecture and Design	21
3.6 Tunneling.....	28
3.7 IPS Container.....	28
3.8 Encryption Devices	31
3.9 Secure Wireless Local Area Networks (SWLANs)	32
3.10 VoSIP.....	35
3.11 VTC.....	38
3.12 Thin/Zero Client.....	41
3.13 IA.....	42
4.0 SIPRNET CONNECTION PROCESS	43
4.1 DISN Connection Approval Process (CAP).....	43
4.2 Accreditation and ATC	44
5.0 COST MODELING AND ESTIMATING FOR FINANCIAL PLANNING.....	46
5.1 General.....	46
5.2 Cost Typing	46
5.3 Funding Requirements and Limitations.....	47

Appendices

Appendix A. Figures, Drawings, and Diagrams.....	A-1
Appendix B. PDS Inspection Checklist	B-1
Appendix C. SIPRNET Site Survey Checklist	C-1
Appendix D. SIPRNET User and Allocation Tables	D-1

Appendices

Appendix E. SIPRNET Allocations for New Military Construction.....	E-1
Appendix F. SIPRNET PDS Specification for MCA Construction.....	F-1
Appendix G. Sample IPS Containers with Movable Racks	G-1
Appendix H. MCA Information System Cost Funding Breakout	H-1
Appendix I. References.....	I-1
Glossary. Abbreviations, Acronyms, and Definitions	Glossary-1

Table

Table 1. VoSIP Area Codes.....	37
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Figures

Figure 1. Epoxy Beads on Connectors.....	21
Figure A-1. Example of WAN, MAN, CAN, and LAN	A-2
Figure A-2. SIPRNET High-Level Overview	A-3
Figure A-3. DISA-Managed SIPRNET Assets	A-4
Figure A-4. NEC-Managed SIPRNET Assets	A-5
Figure A-5. Tenant-Managed SIPRNET Assets	A-6
Figure A-6. SIPRNET Dial-Up Example	A-7
Figure A-7. PDS Physical Architecture Examples	A-8
Figure A-8. Wireless Architectures	A-9
Figure A-9. Secure-Only ISDN or IP VTC	A-10
Figure A-10. Multi-Domain ISDN or IP VTC	A-10
Figure G-1. Hamilton IPS Clearance Requirements.....	G-2
Figure G-2. Hamilton IPS Dimensions.....	G-3
Figure G-3. Trusted Systems IPS Clearance Requirements	G-4
Figure G-4. Trusted Systems IPS Dimensions.....	G-5
Figure H-1. MCA Information System Cost Funding Breakout.....	H-1

Attachment

Attachment 1. Memorandum, U.S. Air Force Judge Advocate, Subject: Medical Use of Encrypted Phone Systems and SIPRNET, 14 April 2004	Att 1-1
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SUMMARY OF CHANGES

Changes for Version 7.0 include the following:

- a. Updated all sections to current regulations and policies.
- b. Updated, replaced, or added appendices and attachments.
- c. Updated references and links.
- d. Included technical criteria for new construction.
- e. Included SRM technical criteria.

SECRET INTERNET PROTOCOL ROUTER NETWORK TECHNICAL IMPLEMENTATION CRITERIA, VERSION 7.0

1.0 INTRODUCTION

1.1 Purpose

The purpose of this document is to provide a set of criteria, culled from numerous Government regulations, policies, and standards into one document, with the intent of assisting a communications service provider, such as the Network Enterprise Center (NEC) on an Army installation, in planning, designing, and installing or expanding SIPRNET capability for its users. This document replaces the *Secret Internet Protocol Router Network (SIPRNET) Technical Implementation Criteria, Version 6.0* and all versions of the *Technical Guide for the Integration of SIPRNET*.

1.2 Scope

This document is not intended to replace any guidance given in any Government regulation, policy, standard, or other official documentation nor to relieve users of the responsibility to ensure their design and installation meets those regulatory requirements. This document incorporates information from the referenced regulations, policies, and standards (see [Appendix I](#)) current as of the time of publication of this document. It is incumbent upon the users of this document to ensure they are familiar with any changes made after the publication of this document to the referenced regulations, policies, and standards. At the same time, a SIPRNET installation that follows the criteria described herein will meet the criteria published in the references and meet accreditation requirements for operational use of the system.

Although this document may contain some information that may be useful in the Outside Continental United States (OCONUS) environment, the exact specifications and requirements in the OCONUS theater may differ from the Continental United States (CONUS) theater. This document is not intended for use as technical criteria for OCONUS implementation. Likewise, this document is not intended for use as technical criteria for classified network implementation in a tactical environment.

1.3 SIPRNET Project Goal

The goal of a typical SIPRNET project is to extend the classified network's Red-side infrastructure to users not collocated with the SIPRNET distribution node or Point of Presence (POP). The methodology used to provide the SIPRNET service will vary based upon such factors as the number of users, their relative locations and concentrations, building layout and construction, operational requirements, and design requirements/limitations imposed by higher headquarters or funding limitations.

Several standard designs have been developed and are discussed in more detail in [Section 3.0](#). These designs are intended for use as a guideline and must be tailored to meet unique requirements at each location.

2.0 MAJOR PARTICIPANTS AND RESPONSIBILITIES

2.1 U.S. Army Corps of Engineer (USACE)

The USACE is responsible for ensuring the construction Contractor selected to install Protected Distribution System (PDS) is technically qualified to do so for the construction projects (Military Construction Army [MCA], Military Construction Army Reserve

SIPRNET Technical Implementation Criteria

September 2013

(MCAR), Sustainment, Restoration, and Modernization [SRM], etc.) it manages. The USACE is also responsible for providing PDS documentation, including shop drawings (i.e., T1, T2, and T4), a List of Materials (LOM), etc., from the construction Contractor to the NEC prior to installation of the PDS with sufficient time (approximately 30 to 60 days) for the U.S. Army Central TEMPEST Technical Authority (USACTTA) to perform a technical review of the PDS. The NEC is responsible for forwarding this documentation to the USACTTA.

2.2 U.S. Army Information Systems Engineering Command (USAISEC)

USAISEC serves as one of the Army's major communications engineering and implementation organizations and offers its services to Department of Defense (DOD) agencies on a reimbursable basis. Services range from performance of site surveys, development of the Engineering Design Plan (EDP) and LOM, procurement of materials, implementation and configuration of systems, and testing of completed installations.

To ensure the collection of all information in an organized and complete manner, USAISEC has adopted the use of a Site Survey Checklist to assist in gathering all required information during the site survey for a SIPRNET expansion project. This document provides a sample checklist, attached in [Appendix C](#), as an example of the information gathered to design a SIPRNET implementation. It is not an all inclusive list, as each site has unique design requirements. To ensure design implementation occurs per all applicable criteria, the surveyor must gather any additional information affecting the design.

2.3 USACTTA

The USACTTA is responsible for performing technical reviews on all PDS designs prior to the procurement of materials or awarding of contracts, determining whether any TEMPEST countermeasures apply and advising the Designated Approving Authority (DAA) on all technical matters pertaining to TEMPEST issues and to a PDS¹ (see [Section 4.2.2](#)).

The request for a PDS technical review must come from a Government official, not a Contractor, prior to committing funds. However, the Government official can designate a Contractor as the Point of Contact (POC) for the review. Technical review requests will be classified as at least Confidential in accordance with (IAW) Army Regulation (AR) 25-2. The PDS information provided in the technical review will be classified as at least Confidential. The same applies to any information relating to approval of a specific PDS. All PDS technical review requests must be sent via SIPRNET e-mail to the TEMPEST Program Manager (TPM) at 902d310thTEMPEST@mi.army.smil.mil.

The request for a PDS technical review must include the information required by National Security Telecommunications and Information Systems Security Instruction (NSTISSI) 7003, Annex C and will be classified appropriately, based on its accumulative content. The USACTTA technical review determines whether the PDS design is compliant with NSTISSI 7003 and will prepare a PDS technical review memorandum for the DAA.

¹ AR 380-27, Chapter 3.

If the USACTTA technical review finds the proposed PDS design is non-compliant, they will assist the customer to develop a compliant design. If the design cannot be made fully compliant, the USACTTA will provide a risk assessment of the proposed design to the DAA.

All Army PDSs will be approved according to NSTISSI 7003, AR 25-2, and AR 380-27.

A copy of the PDS technical review must be included as part of the PDS accreditation package for review/approval by the DAA.

The DAA has the authority to accept the risk of a non-compliant PDS and must formally accept any risk in writing. A copy of the USACTTA risk assessment must be attached. A copy of the risk acceptance memorandum must be provided to the supporting USACTTA².

2.4 DAA

The DAA is responsible for ensuring all Information Systems (ISs) are properly certified and accredited. The DAA is also responsible for approving all local connections to the Campus Area Network (CAN) at the site.

The DAA is vested with the authority to formally assume responsibility for operating an IS at an acceptable level of risk. The DAA must weigh the operational need for the systems capabilities, the protection of personal privacy, the protection of the information being processed, and the protection of the information environment.

The DAA legally accepts any unmitigated risk associated with the approval of the system.

2.5 NEC/Communications Service Provider

The NEC/communications service provider is responsible for the following:

- a. Obtaining the systems' DOD Information Assurance Certification and Accreditation Process (DIACAP) accreditation for all ISs that connect to the installation SIPRNET.
- b. Obtaining PDS technical reviews from the USACTTA.
- c. Providing the required Red-side and Black-side Internet Protocol (IP) addresses. This includes obtaining additional IP subnets to satisfy new requirements.
- d. Configuring the CAN to pass encrypted, classified network service to remote buildings, as required, if such service is tunneled through the CAN.
- e. Ensuring the Black-side connectivity through the facility's Local Area Network (LAN) and the post CAN is available for all encryption devices, as required.
- f. Implementing any new circuits required for the installation or expansion of a classified network service on the installation.
- g. Procuring encryption devices through the Communications Security Logistics Activity (CSLA) and coordination with the local Communications Security (COMSEC) custodian for COMSEC devices that fall within their control.

² AR 380-27, Chapter 3.

SIPRNET Technical Implementation Criteria

September 2013

h. Coordinating with the COMSEC Custodian for the procurement and issuing of encryption key material and key material loading devices.

The Information Assurance (IA) Manager (IAM) is responsible for the overall security of the network. These items will include security oversight, vulnerability assessments, and IA Vulnerability Alert (IAVA) compliance roll-ups, as well as other areas.

In conjunction with the user, the Information Management Officer (IMO) is responsible for the procurement, installation, and configuration of application software to be installed on end-user computer resources.

Coordination with the Directorate of Public Works (DPW) will take place to complete any agreements regarding site preparation issues (space, power, outlet locations, etc.).

The communications service provider will perform these responsibilities on military installations that do not have a NEC.

2.6 User IMO

User IMO is responsible for the following:

- a. Maintaining the physical and information security of the classified network access.
- b. Providing information to the NEC for system accreditation.
- c. Ensuring user equipment is properly marked with the appropriate classification level for the classified network on which it is used.
- d. Ensuring user equipment is stored in approved security containers or in an open storage area during non-duty hours.
- e. Procuring COMSEC devices (IAW organizational procedures) through CSLA, and coordinating with the COMSEC custodian for COMSEC material within their control.

2.7 Shared

The NEC, IMO, and user must coordinate the extension of the SIPRNet drops, ordering of equipment, and overall accountability for the system.

The NEC, IMO, and user must coordinate closely with each other to ensure the implementation of all physical security and operating procedures required for the system to be accredited.

The construction Contractor shall provide PDS documentation (shop drawings, LOM, etc.) to USACE. USACE will forward the PDS documentation to the NEC for submission to the USACTTA prior to the installation of the PDS. The NEC will forward this documentation to the USACTTA for technical review. USACE will also ensure the construction Contractor is technically qualified to install a PDS.

Do not commit funds for the acquisition of material or the installation of a PDS until after the USACTTA has completed its technical review of the PDS design and its comments (if any) are addressed.

3.0 TECHNICAL SOLUTIONS DESCRIPTION AND CRITERIA

3.1 Construction Criteria

3.1.1 General

There have been several programs in the past focused on expanding the SIPRNET user base. Over time, many lessons have been learned, and those lessons have been incorporated into these criteria.

Two remaining programs, MCA and SRM, are still relevant and are discussed in subsequent sections of this document. The most prevalent SIPRNET implementation is the organization/agency-funded implementation. These projects vary in size from a few to several hundred new connections.

3.1.2 MCA

SIPRNET distribution will be installed in most, but not all, buildings (see [Appendix E](#)) and originate from a secure Telecommunications Room (TR) (see [Section 3.5.5](#)).

All private offices and conference rooms, not to exceed 25 percent of designed building occupancy, shall be designed to have SIPRNET capability. A SIPRNET-capable room requires either non-operational or lockable windows; Heating, Ventilation, and Air Conditioning (HVAC); and sound attenuation to at least Sound Transmission Category (STC)-35 whether or not a SIPRNET drop is installed in the space.

For example, a new building designed for a 100-person occupancy that has 20 private offices and conference rooms is authorized 25 SIPRNET User Drop Boxes (UDBs). In this scenario, each office would receive one SIPRNET UDB with one cable connection; however, conference rooms and Commander/Director offices would receive one UDB with two cable connections. To accommodate occasional SIPRNET users, five UDBs with one cable connection are located in a room designed and designated as a SIPRNET café. Design and use of SIPRNET cafés (see [Section 3.5.8](#)) is highly encouraged to accommodate the occasional SIPRNET user group.

Conference rooms using amplified sound (e.g., secure Video Teleconference [VTC]) during a classified meeting may require sound attenuation of at least STC-35 or more for the walls, ceiling, and door(s). Contact local security (physical, information, etc.) to determine conference room sound attenuation requirements.

SIPRNET drops located in buildings with secure rooms (e.g., Network Operations Center [NOC], Brigade Operations Center [BOC], Division Operations Center [DOC], Joint Operations Center [JOC], etc.), Sensitive Compartmented Information Facilities (SCIFs), or secure buildings should not count against the 25-percent occupancy rule.

MCA projects are funded for SIPRNET installation to the battalion level and higher. SIPRNET installation exceptions below battalion level include Explosive Ordinance Detachment (EOD), Military Intelligence (MI), and Special Operations Forces (SOF) companies and platoon/detachment facilities. Space for secure SIPRNET TRs will be designed and installed for all facilities authorized for SIPRNET, as well as for Company Operating Facility (COF) and Tactical Equipment Maintenance Facility (TEMF) for future use.

SIPRNET Technical Implementation Criteria

September 2013

3.1.3 General Guidance for Building Rehabilitation and Retrofit – SRM

Numerous profiles for the installation of SIPRNET in existing facilities are included in the tables provided in [Appendix D](#) for use as general guidance in design and estimation. The quantities in the tables reflect the guidelines from the Mission Command (MC) Networks and Systems/Network Operations Service List.

For buildings that do not fit into the profiles in [Appendix D](#), up to five SIPRNET drops will be engineered and installed for those buildings accommodating personnel in the grade/rank of O-5/Lieutenant Colonel and above (or the civilian equivalent) with command responsibilities.

Use the same design criteria specified in [Section 3.1.2](#) when programming funding for and designing of SIPRNET access in SRM projects.

3.1.4 Medical Facilities

Hospitals and medical facilities, whether existing or new construction, will be provided only one SIPRNET drop per building due to Geneva Convention and Law of Armed Conflict (LOAC) issues (see Category 510 in [Appendix E](#)).

The SIPRNET access in medical facilities may not be used for Command and Control (C2) or for intelligence gathering. SIPRNET use will be limited to the exclusive purpose of retrieving and transmitting medical information³.

3.2 Connectivity Areas of Responsibility

3.2.1 General

SIPRNET implementation and access spans from the individual user up to the global SIPRNET Wide Area Network (WAN). The responsibility for operating, maintaining, and implementing such access is divided into three main areas of responsibility, each by a different organization (i.e., Defense Information Systems Agency [DISA], NEC, and tenant/user).

The areas of responsibility between the following three agencies correspond roughly to the size of the network: WAN, Metropolitan Area Network (MAN)/CAN, LAN. Refer to [Figure A-1](#) for a depiction of the various network sizes and to learn how these networks interrelate.

Each of the three areas of responsibility must be addressed for a successful implementation of SIPRNET access. [Figure A-2](#) depicts a high-level view of the overall user-to-WAN reach of the SIPRNET, with the three areas of responsibility delineated.

3.2.2 DISA-Managed

[Figure A-3](#) shows an expanded view of the DISA-managed portion of the SIPRNET. This drawing represents a composite of multiple design options and will not apply to every situation. This drawing is intended to give the reader a conceptual view of various configurations, architectures, and security designs, as described in the ARs, Security Technical Implementation Guides (STIGs), and other policy documents.

³ Memorandum, U.S. Air Force Judge Advocate, 14 April 2004.

As DISA manages and controls the circuit bringing SIPRNET access into the post, DISA also manages all of the equipment associated with the circuit. Although controlled by DISA, the near-end equipment used to terminate the circuit at the post is located in one of the NEC facilities on post. [Figure A-3](#) does not show an all-inclusive list of equipment, but only a representation of the types of equipment.

As indicated in [Figure A-3](#), SIPRNET traffic is delivered from the DISA Global Information Grid (GIG), through COMSEC to the NEC SIPRNET POP using a variety of transmission methods, ranging from point-to-point serial circuits to fiber optic Synchronous Optical Network (SONET) channels. When providing SIPRNET connectivity to a post, DISA will provide the proper circuit for the bandwidth requested. The requesting agency in this case, the NEC, must fund all costs associated with the provisioning of the circuit, including equipment and labor costs for equipment installed at both ends of the circuit. In lieu of providing the funding, agency requesting the circuit may elect to provide the equipment directly to DISA and the equipment must be that specified by DISA during the circuit engineering process.⁴ Funding must still be provided to DISA for the installation-related costs.

The circuit conditioning equipment and encryption device are typically located on the post in a NEC-controlled facility, even though they are DISA assets. The demarcation line between the DISA and NEC responsibilities is typically the Red-side connection on the encryption device used between the SIPRNET WAN and the NEC-controlled equipment.

3.2.3 NEC-Managed

[Figure A-4](#) shows an expanded view of the NEC-managed portion of its SIPRNET connectivity. This drawing represents a composite of multiple design options and will not apply to every situation. This drawing is intended to give the reader a conceptual view of various configurations, architectures, and security designs, as described in the ARs, STIGs, and other policy documents.

As indicated in [Figure A-4](#), between the Red-side of the DISA-controlled encryptor and the SIPRNET POP distribution router, there is a suite of network security equipment. The local NEC will determine the actual type, quantity, and configuration of the equipment (per DISA policy⁵), as it is responsible for the network security of everything behind the Red-side of the encryption device (termed an “enclave” from the WAN perspective of DISA). Collectively, the routers, firewalls, and other equipment are referred to as the NEC SIPRNET POP. The extension of SIPRNET service to tenants/users from the SIPRNET POP is the main concern of the majority of SIPRNET expansion projects and this document. The following lists four basic scenarios for these SIPRNET expansion projects:

- a. **Scenario 1** - The first and most common scenario is the military installation where a NEC (or other service provider) has a SIPRNET POP and uses the existing CAN for extending the service to the tenant/user that is also located on the military installation.

⁴ DISA Memorandum, Process for Use of Site Support Task, Enclosure 1, Paragraph 4.

⁵ DISA Network Infrastructure STIG, Section 2.1.

SIPRNET Technical Implementation Criteria

September 2013

Although this scenario will cover the majority of implementations, other solutions may be required to meet the tenants' exact mission needs.

b. **Scenario 2** - In the second scenario, a remote U.S. military or Government user that is not on the military installation needs SIPRNET service from an existing SIPRNET POP on the military installation (i.e., an off-post extension). Typically, this scenario involves a small user group (1-20 users), but may be employed for larger groups. To implement this scenario, coordinate the physical security requirements of the SIPRNET network equipment at the remote user location with the SIPRNET service provider, such as the NEC on the military installation where the circuit will originate. Although this scenario involves a local, off-post circuit extension under the control of the NEC, it must be coordinated with the DISA Connection Approval Office (CAO)⁶ prior to implementation. When DISA approves the extension of this service, the originating POP (the NEC) provides and controls the address space used at the remote user location and is responsible for accounting for the certification of this new location (documenting the network configuration). Note the use of private IP address space (10.0.0.0, 172.16.0.0, 192.168.0.0, etc.) is not allowed on SIPRNET networks.⁷

c. **Scenario 3** - In the third scenario, the user requires SIPRNET access in a non-military environment, such as in an off-post Contractor's facility. This scenario is similar to the previous scenario except for the affiliation of the remote user. Again, the user must address the physical security requirements for the SIPRNET network equipment that is located at the user's location. The type of network security equipment required must be coordinated with the SIPRNET service provider, such as the NEC on the military installation where the circuit originates. As the remote user is not a military or Government unit, a Government agency must sponsor the SIPRNET connection.⁸

d. **Scenario 4** - In the fourth scenario, the user needs a SIPRNET dial-up capability for limited use by a very small group of users. Dial-up SIPRNET is an inexpensive means to provide SIPRNET access to those who require limited service using commercial telephone lines. Users gain access by dialing a DISA communications server using a Secure Data Device (SDD)-1910 or Secure Terminal Equipment (STE). If dial-up service is required, the user should coordinate directly with DISA. Dial-up access is restricted to Government or military personnel only and is not available for Contractor personnel. [Figure A-6](#) depicts a simplified account of a dial-up connection from the tenant to the DISA server.

3.2.3.1 Post/Station Transport

Although encryption devices will be required in most situations, connectivity between the NEC SIPRNET POP and the remote user location may be provided in various ways. The following paragraphs note three of the more prevalent methods of connectivity:

⁶ DISA Network Infrastructure STIG, Section 2.3.

⁷ DISA Network Infrastructure STIG, Section 2.8.

⁸ DISN Connection Process Guide, Paragraph 2.1 and Enclosure C.

a. **Alternative 1** – Traversing the existing CAN. The SIPRNET circuit travels from the NEC POP to each End User Building (EUB) over the existing CAN infrastructure in encrypted (Black) form. This is referred to as tunneling, and this is discussed in more detail in [Section 3.6](#). Traversing the existing CAN is the preferred method of inter-building transport.

b. **Alternative 2** – Using a dedicated cable path. Dedicated cable pairs, either copper or fiber, may be used to extend SIPRNET service in the encrypted (Black) form. This method does not use the post cable infrastructure in an efficient manner, so it is not the preferred method. As the information is encrypted, the use of a PDS is not required.

c. **Alternative 3** – Using an exterior PDS. As the information being transmitted is not encrypted (it is Red or plain text), a PDS is required. An exterior PDS⁹ between buildings is an expensive method for inter-building transport, especially if there are multiple buildings, or the buildings are not immediately adjacent to each other. The SIPRNET service is extended in unencrypted (Red) form, eliminating the cost of the COMSEC devices; however, the PDS cost may more than offset the COMSEC cost. In addition, physical security checks and procedures are required to ensure the exterior PDS remains secure. For these reasons, an exterior, hardened carrier PDS is not a preferred method of transport between buildings.

3.2.4 Tenant/Local/User-Managed

[Figure A-5](#) shows an expanded tenant/user-managed portion of the SIPRNET connection drawing. This drawing represents a composite of multiple design options and will not apply to every situation. This drawing is intended to give a conceptual view of various configurations, architectures, and security designs, as described in the NSTISSIs, ARs, STIGs, and other policy documents.

A port on each building's unclassified network switch (EUB switch) is reserved for the encrypted (Black) side of the encryption device. Connect the Black side of the encryption device to the reserved switch port using either a fiber optic or Ethernet cable. If the encryption device is located in a room other than the TR, such as an office, the encryptor cable can use the existing building cable distribution infrastructure (i.e., cable trays) to the TR to connect the reserved switch port.

The Red tenant router is optional equipment and is only required if a Virtual Private Network (VPN) is required to pass traffic between two Red enclaves separated by a pair of Inline Network Encryptors (INEs) and the traffic is of a type that does not get passed by the INE.

Distribution of the SIPRNET service from the switch to the users normally occurs via a PDS that is determined by the physical locations it must traverse. [Section 3.4](#) discusses this subject in detail.

⁹ NSTISSI 7003, Annex B, Paragraphs 4a(1)(d) and (e).

SIPRNET Technical Implementation Criteria

September 2013

3.3 Access Areas

3.3.1 General

There are three types of access areas: Controlled Access Area (CAA), Limited Controlled Area (LCA), and Uncontrolled Access Area (UAA). The type of access area and the local threat environment determine the type of PDS and other supplemental security measures required to protect the classified information transiting the area¹⁰.

Local threat levels are designated as low, medium, or high. The exact designation of the threat level for each specific geographical area is contained in a classified National Security Agency (NSA) document (*Information Assurance Technical Capabilities Report*) and is beyond the scope of this unclassified document.

3.3.2 UAA

A UAA is an area over which no personnel access controls are or can be exercised.¹¹ In short, it is an area that is open to the public. There are no personnel controls that ensure only authorized personnel are allowed into the area or those personnel entering the area have a security clearance. This is the least secure of the types of access areas.

A hardened carrier PDS is required for the transmission of all unencrypted, classified information through a UAA at any threat level.

3.3.3 LCA

An LCA is an area with a PDS traversing through it where exploitation of the PDS is not considered likely or where legal authority to identify and remove a potential exploitation exists.¹²

If the area has some personnel access controls (i.e., it is not a UAA), yet does not meet the definition of a CAA, then it is an LCA. A Government or military facility with a locked door where a visitor, once admitted to the facility, can roam unescorted is an example of an LCA. A hardened carrier PDS is required for the transmission of all unencrypted Secret or higher classified information through an LCA at any threat level.

3.3.4 CAA

A CAA is an area in a facility that is under direct physical control within which unauthorized persons are denied unrestricted access. If unauthorized personnel are granted access, they are escorted by authorized persons or are under some form of continuous physical or electronic surveillance.¹³

A CAA does not automatically qualify for or may not be rated for the open storage of classified material.

Use a simple carrier PDS in lieu of a hardened carrier PDS in a CAA. The distribution system for unencrypted SIPRNET within a Secret CAA may be constructed of the same

¹⁰ NSTISSI 7003, Annex B, Table B-1.

¹¹ NSTISSI 7003, Annex A Paragraph g.

¹² NSTISSI 7003, Annex A Paragraph d.

¹³ NSTISSI 7003, Annex A Paragraph b; AR380-27.

materials as a simple PDS for unencrypted SIPRNET in a Confidential CAA, but it is not a PDS; therefore, the additional inspection requirements for a PDS do not apply. Coordinate physical security and procedural issues with the proper, local physical and information security personnel.

The U.S. Army Network Enterprise Technology Command (NETCOM) has specified CAA construction requirements in Paragraph 38 of NETCOM Regulation 380-5 for CAAs that fall under its jurisdiction. Consult with local physical, information, and G2 security personnel and the cognizant Commander to determine CAA accreditation requirements.

3.3.5 Open Storage Area

An open storage area is a secure room or vault that meets the physical security standard specified in Paragraphs 7-13 and 7-14 of AR 380-5 and is accredited by the cognizant physical security office and Commander.

A PDS is not required inside an open storage area for classified data cabling at or below the security level of the open storage area. Use standard cable distribution methods and maintain Red/Black cable separations, as established by the supporting USACTTA.

The preferred method of safeguarding classified material is to secure it inside a General Services Administration (GSA)-approved safe. Designation of an open storage area will only be approved when storage in a GSA-approved safe is not feasible due to size, shape, or volume of the material stored.¹⁴ In short, the use of open storage areas is not the preferred method of securing classified material.

The Commander is responsible for designating an area as restricted or controlled to safeguard property or resources, such as the classified material or information, for which the Commander is responsible.¹⁵ To be designated as an open storage area, the selected area must be designated as such by the Commander and meet the appropriate physical security standards. A physical security inspection and/or a security engineering survey is required as part of the approval process.¹⁶ Upon successful completion of the inspection and recommendation from the physical security personnel, the Commander may designate in writing that an area is approved for open storage at a given classification level.

The physical security construction standards for a secure room, open storage area at the Secret level are as follows:¹⁷

- a. The floor, walls, and ceiling shall be made of permanent construction materials that offer resistance to, and evidence of, unauthorized entry into the area. In addition, the walls shall extend from the true floor to the true ceiling. The wall extensions may also be made with 18-gauge wire mesh or expanded steel screen, if the permanent walls do not extend from true floor to true ceiling.

¹⁴ AR 380-5, Paragraph 7-12.

¹⁵ DOD Reg 5200.08-R, Paragraph C3.2.4 and AR 190-13, Paragraph 6-3.

¹⁶ FM 3-19.30, Chapter 11.

¹⁷ AR 380-5, Section 7-13 Paragraph b and Section 7-20.

SIPRNET Technical Implementation Criteria

September 2013

b. The doors to the area shall be substantially constructed of wood or metal. Wood doors shall be solid throughout. The hinge pins of doors that swing outward from the area (i.e., the hinges are accessible from outside the area) shall be pinned, brazed, or spot-welded to prevent their removal.

c. The access door into the area shall be equipped with a built-in combination lock, such as the CDX-09.¹⁸ CDX-07 and CDX-08 lock parts are no longer manufactured; therefore, these locks should not be installed on new installations. Doors, other than the access door, shall be secured from inside the area to prevent entry from outside. Key-operated locks that can be accessed from outside the area are not authorized.

d. If there are windows in the secure area where the window's bottom is less than 18 feet from the ground or that are easily accessible from a nearby object (such as a tree or fence), then the windows shall be covered with material to help prevent forced entry and raise the physical strength to a level at least equivalent to the surrounding walls. Typically, metal bars or wire mesh screening is used to accomplish this. The method of securing the covering over the windows is part of the security system, so it shall also provide a protection level at least equivalent to the walls. In short, the fasteners shall be inaccessible from outside or be non-removable.

e. Windows which might reasonably afford a view of the classified information inside the area shall be made opaque or be equipped with blinds, drapes, or other coverings.¹⁹

f. Any openings into the room, such as for air conditioning ducts or vents, are required to be less than 96 square inches. If larger than 96 square inches, the opening shall be hardened to resist penetration²⁰. Openings that exceed 96 square inches and have at least one dimension that does not exceed 6 inches do not require hardening. The basic method used to harden the opening is to break up the opening into a number of smaller openings, rather than cover it over or just reduce the size.²¹

g. An Intrusion Detection System (IDS) (i.e., Integrated Commercial Intrusion Detection System [ICIDS] alarm system) is required.²²

h. Physical security construction requirements vary from one location to another and are more stringent at OCONUS locations. Contact the local physical security and G2 offices to determine the threat level and physical security construction requirements that pertain to a specific location.

¹⁸ Fed Spec FF-L-2740A with Amendment 1 and Federal Qualified Products List QPL-FF-L-2740-8.

¹⁹ AR 380-5, Section 7-20, Paragraph d(1).

²⁰ AR 380-5, Section 7-13, Paragraph b(5).

²¹ MIL-HDBK-1013/1A, Paragraph 5.6.7.

²² AR 380-5, Section 7-20 Paragraph e and Section 7-14.

3.4 PDS

3.4.1 General

A PDS carrier is used to distribute cabling that transmits unencrypted (plain text), classified information through an area of lesser classification or control.²³ The type of PDS required is determined by the type of access area the PDS transmits through, as well as the local threat level. The less secure the access area is and the higher the local threat level, the more physical security must be provided by the PDS to the classified cable inside the PDS.²⁴ Since PDSs inherently offer greater levels of risk, the use of PDSs should be minimized in Army facilities and systems. A PDS is not recommended in high threat areas.²⁵

The PDS begins at the point where the carrier leaves the Information Processing System (IPS) container, secure TR, or secure room that houses the building POP and ends at the surface-mounted UDB the user unlocks and opens to access network connections. The portion of the PDS in the middle, the part that carries the classified cabling from one area to another, is known as the carrier.

3.4.2 PDS Purpose

The PDS must provide adequate electrical, electromagnetic, and physical safeguards to deter exploitation. Since PDS is penetrable, given the opportunity and adequate time, a philosophy of detection of attempted penetration is employed.²⁶

3.4.3 Simple Carrier PDS

A simple PDS is used for SIPRNET cable distribution within a Confidential rated CAA. A PDS is not required in a Secret or higher rated CAA.²⁷

In a simple PDS, the cables must be in a carrier constructed of any material (e.g., wood, Polyvinyl Chloride [PVC], Electrical Metallic Tubing [EMT], etc.). However, the joints and access points shall be controlled by personnel cleared to the highest level of data handled by the PDS.²⁸

The most common carrier used in a CAA is a surface-mounted plastic raceway or subfloor cable tray distribution system.

Use the standard cable management/distribution system to provide an orderly scheme to route the cables, maintain Red/Black cable separation, and maintain separation from power cables.²⁹

²³ NSTISSI 7003, Annex A, Paragraph e.

²⁴ NSTISSI 7003, Table B-1.

²⁵ AR 380-27, Section 3-1.

²⁶ NSTISSI 7003, Page 3, Section IV, Paragraph 5.

²⁷ NSTISSI 7003, Annex B, Table B-1.

²⁸ NSTISSI 7003, Annex B, Paragraph 4.b(2).

²⁹ NSTISSAM 2-95, Para 4.5.

3.4.4 Hardened Carrier PDS

3.4.4.1 General

A hardened PDS is required for SIPRNET cabling where the PDS transmits an [LCA](#) or a [UAA](#). There are three types of hardened PDSs: hardened carrier, continuously viewed carrier, and alarmed carrier. The following sections discuss these types of PDSs.

3.4.4.2 Hardened Carrier

A hardened carrier is the most common type of PDS. It is so common, the term “hardened PDS” or just “PDS” is often mistakenly used in lieu of the term “hardened carrier.” The hardened carrier shall be a ferrous metallic conduit or duct. The fittings and connectors shall be made of the same ferrous metal material. All seams and joints of the PDS shall be permanently sealed completely around all surfaces by either welding (continuous or tack) or two part epoxy. High compression fittings (i.e., couplers, box-connectors, etc.) shall be used and sealed with epoxy applied completely around all mating surfaces. The covers of all access points, such as pull boxes (and conduit body covers), shall be sealed likewise (welded or epoxy), unless they are secured with a GSA-approved, changeable combination padlock. Boxes with pre-punched knockouts are not acceptable and shall not be used in any PDS.³⁰ The only changeable combination padlock currently approved by GSA is the Sargent & Greenleaf model 8077AD.³¹

The hardened carrier most personnel are familiar with is EMT conduit. When using EMT conduit, the fittings, conduit bodies, and connectors shall be made of the same ferrous metal. The use of cheaper “pot metal” and aluminum-type fittings and conduit bodies are not permitted. Non-metallic fittings shall not be used in any hardened carrier system except when mandated by the USACTTA. A non-conductive break may be required for TEMPEST reasons for some PDSs.

Conduit is defined as a tube, duct, or protected trough used for carrying and routing wires or cabling. Conduit may be round, square, or rectangular. It most commonly is a closed system with only the ends being open for pulling cables through.

Secure raceway is a type of conduit with a removable cover (top-cap), enabling cabling to be laid into the conduit rather than pulled through the conduit. The use of a hardened carrier with a removable top-cap cover allows for the future addition of extra cables and expansion of the PDS (referred to as “future adds, moves, and changes”) with minimal effort. This benefit is offset by the requirement to remove the sealing material from the carrier, which may also damage the carrier and require its replacement at added expense. For raceway with a removable cover, the requirement to seal all joints and edges applies to the entire length of the removable cover, as well as to the edges and joints of the fittings. The extra labor involved in sealing both sides of a removable cover for the length of the entire hardened carrier makes this a more expensive solution than standard conduit, such as EMT.

³⁰ NSTISSI 7003, Annex B, Paragraph 4.a(1).

³¹ E-mail, GSA, Mr. Christopher Pollock, 4 May 2010.

The DAA can choose to accept the risk of a carrier that has not been permanently sealed with epoxy or another approved method. The epoxy waiver previously granted by the 7th Signal Command has been rescinded. The DAA must sign a risk acceptance for each non-compliant PDS. The risk acceptance must reference the USACTTA site-specific threat assessment.

Note1: Local physical security personnel may, at their discretion, override this risk acceptance and require the use of epoxy to seal the secure raceway. It is always an option to increase the security measures.

The conditions for obtaining a risk acceptance from the DAA for permission not to permanently seal the PDS are as follows:

- a. The hardened carrier used shall be a secure raceway with a removable top-cap or raceway cover manufactured from ferrous material.
- b. The removable top-cap or raceway cover shall be the only portion of the secure raceway to which the site epoxy waiver applies. Remaining components of the PDS carrier shall have epoxy applied.
- c. Technicians certified by Holocom Networks must install secure raceway manufactured by Holocom Networks.
- d. Legrand Wiremold Data-Fence secure raceway must be installed IAW the manufacturer's instructions.
- e. Legrand Wiremold Data-Fence, secure channel (4 3/4" x 1 3/4", 2" x 2" or 1" x 1") raceway components and fittings will have epoxy applied at all mating surfaces.
- f. Although hardened carrier typically is used inside a building or facility, it may also be used to carry classified cabling between buildings. This is usually a cost prohibitive solution, especially when the PDS must be run between multiple buildings in a point-to-multipoint configuration, such as from an NEC facility to multiple tenant buildings on a military installation. Hardened carrier installed between buildings may be buried or suspended.

Note 2: All horizontal and work area outlet cable distribution within a building/facility routed in a conduit buried beneath a building/facility foundation or routed in conduit within a building/facility foundation or floor slab must be reviewed by the USACTTA and approved by the DAA. The USACTTA review and DAA approval must take place before the building/facility design is complete or issued for new construction (MCA, MCAR, SOF, etc.) projects, SRM projects, or extending SIPRNET access within existing facilities.

- g. If buried, the hardened carrier shall be at least 1 Meter (m) below the surface on property owned or leased by the Government. If manholes are used in the carrier distribution system, they shall be secured using a GSA-approved, changeable combination padlock or shall be welded closed. These requirements are for the lower threat

SIPRNET Technical Implementation Criteria

September 2013

environments located in CONUS only. For higher threat environments, the buried, hardened carrier shall also be encased in at least 8 inches of concrete.³²

h. If the hardened carrier is to be suspended between buildings, it shall be a minimum of 5 m above the surface grade level, and the area the PDS traverses over shall be owned or leased by the Government. In addition, the area containing the PDS shall be illuminated to help deter tampering.³³

3.4.4.3 Continuously Viewed Carrier

A continuously viewed carrier is a PDS carrier that is under direct, continuous observation 24 hours per day. There must be operational procedures in place to ensure all attempts to disturb the PDS are investigated by security personnel within 15 minutes.³⁴

There is no restriction on the type of carrier, so it may be made of any material and does not need to be sealed, as it is a guarded system. The security afforded the classified cabling stems from the quick response by security personnel to any attempt to compromise the carrier.

3.4.4.4 Alarmed Carrier

An alarmed carrier is a carrier protected by an alarm system that is approved by the cognizant COMSEC and/or physical security authorities.³⁵ The use of an alarmed carrier as a hardened PDS, therefore, must be closely coordinated with these authorities from the design phase through the installation and certification phases.

Consider the following areas when designing an alarmed carrier PDS:

- Physical topology of the alarm loop circuit from the Alarm Processing Unit (APU) to the monitoring location
- Point-to-point monitoring is the monitoring of a single point (i.e., UDB or backbone cable run between a secure TR)
- Point-to-point or point-to-multipoint alarm circuits
- Point-to-point monitoring is monitoring a single cable alarm circuit
- Point-to-multipoint monitoring is monitoring multiple cables (quantity is dependent on APU) on a single alarm circuit loop
- Verification of performance per Table B-4, NSTISSI 7003
- Procedures when alarm condition occurs
- Alarm management - What organization (Program Management Office [PMO], NEC, NOC, etc.) is responsible for monitoring APU alarms?
- Alarm notification method (e-mail, Security Management Server [SMS], Simple Network Management Protocol [SNMP] network management suite, etc.)

³² NSTISSI 7003, Annex B, Paragraph 4.a(1)(d).

³³ NSTISSI 7003, Annex B, Paragraph 4.a(e).

³⁴ NSTISSI 7003, Annex B, Paragraph 4.a(3).

³⁵ NSTISSI 7003, Annex B Paragraph 4.a(2).

- Alarm response by security personnel within 15 minutes
- Actions taken to terminate data transmission over the cabling effected by the alarm
- Procedures to clear alarm conditions
- Inspection of all cables in the alarm loop circuit
- Investigation of intrusion attempt

An example of an alarmed carrier protected by an internal alarm system would be an electronic system used to monitor and report on the status of the cables inside the carrier. This type of system monitors the physical layer (Open Systems Interconnection [OSI] Model Layer 1) with a fiber optic cable detecting variations in the timing and shape of the light pulses caused by vibrations because of an attempted penetration using unused, dark strands of cable. By operating at the physical layer, the alarm system does not capture, process, or record any classified data in the cable. Two examples of this type of fiber optic, alarmed carrier are the Interceptor, manufactured by Network Integrity Systems, and the SecureLAN by Fiber SenSys (additional details are available at www.fibersensys.com, www.gocsc.com/secureIT, or www.networkintegritysystems.com).

3.4.5 PDS Inspections

3.4.5.1 Periodic

During its operational lifetime, a PDS must periodically be inspected physically and visually for signs of penetration, tampering, or degradation.³⁶ For a Secret PDS transmitting a UAA or an LCA within a lower threat environment in the U.S., the visual inspection shall be at least once per day on a random schedule. This also applies to a Secret-level (i.e., SIPRNET) PDS that traverses a Confidential CAA. Refer to Annex B of NSTISSI 7003 for the PDS Inspection Matrix and Schedules.³⁷

To aid in the conduct of the visual inspection, the PDS shall not be installed in a concealed location such as inside walls, above suspended ceilings, or below raised flooring.

Daily inspection of the PDS includes non-duty days, not just normal duty days. Service providers and users must determine the policy they will use to conduct daily inspections of the PDS on non-duty days for facilities not occupied by personnel on non-duty days. One random technical inspection is required annually. The technical inspector shall be sufficiently trained to recognize changes in the technical aspects of the PDS (e.g., by-pass circuitry, attachment or removal of devices or components, etc.), inappropriate or suspicious signal levels, mechanical changes, and the TEMPEST integrity of the PDS.

3.4.5.2 Post-Installation

Inspect the PDS after installation to ensure it meets all applicable installation guidance and regulations. [Appendix B](#) has a link to a sample Quality Assurance (QA)/Quality Control (QC) inspection checklist to assist in the PDS inspection process. Modify the list, as

³⁶ NSTISSI 7003, Annex B, Paragraph 1.a(6).

³⁷ NSTISSI 7003, Annex B, Table B-2.

SIPRNET Technical Implementation Criteria

September 2013

required, to accommodate the specific installation requirements. The inspection should take place in the following two parts:

a. **Part One – Inspection before Epoxy Application:** Perform this inspection before the application of the epoxy to the PDS. This will permit the inspector to inspect the junction and pull box interiors. It also allows the inspector to check for the proper installation of the locking, removable top-cap duct to ensure the locking mechanisms are properly installed and functioning.

b. **Part Two – Inspection after Epoxy Application:** Perform this inspection after epoxy is applied to the PDS and has hardened. Its main purpose is to ensure the epoxy has been properly applied. This part of the inspection is also the ideal time to check for finishing work, such as the appearance of the PDS and the work site.

Additional inspections after the PDS carrier is installed, but before cabling is installed, are highly recommended.

3.4.6 PDS Carrier

The PDS is comprised of two basic components that make up the entire system: the [PDS carrier](#) (conduit, duct, or raceway) and the [UDB](#). Only Red (unencrypted, classified) SIPRNET cabling shall be installed inside the PDS carrier. Ethernet cable contained in the PDS carrier that passes National Security Information (NSI) traffic shall have a red-colored cable sheath.

The PDS installation shall meet NSTISSI 7003 standards, rather than the National Fire Protection Association (NFPA) 70 or Building Industry Consulting Services International (BICSI)/Telecommunications Industry Association (TIA) standards. For the installation of a PDS, this document and other pertinent communications policies and regulations, most notably NSTISSI 7003, shall take precedence. In all other instances, such as the installation of conduits housing Alternating Current (AC) power, the policies and guidance in NFPA 70 must take precedence. The following requirements apply to a PDS installation:

- a. Properly trained and equipped personnel shall install the PDS.
- b. All conduit, duct, or raceway (generically referred to as “PDS carrier”) shall be installed level and plumb wherever possible along its entire length.
- c. The PDS carrier shall be installed using spacers or other standoff mounting hardware (a minimum of ½ of an inch) to allow for visual inspection of all sides of the PDS carrier.
- d. Cable fill shall not exceed 80 percent for removable cover PDSs and 60 percent for solid tube or conduit PDSs.
- e. A pull string shall be installed in each run between pull points or pull boxes in solid construction and EMT PDS carriers and shall be replaced each time cable is pulled through.
- f. For a solid construction PDS carrier, a pull point or pull box shall be installed at least every 180 degrees of bend or change in direction. Pull and junction boxes may be mounted flat against the mounting surface. Conduit bodies may be used in lieu of pull boxes, as pull points in EMT carriers.
- g. The 1-inch conduit body (LB, LL, LR, T, X, etc.) is the smallest size conduit body used in a PDS to maintain a minimum, no-load cable bend radius of 1 inch for Ethernet cable and 2 inches for fiber optic cable.

- h. The PDS carrier shall be securely fastened to the mounting surface at least every 5 feet of length and within 18 inches of each end of a carrier section to the greatest extent possible.
- i. The PDS carrier shall be mounted with a minimum of 4 inches of clearance between the top of the carrier and the ceiling to allow for proper sealing of the PDS; however, a clearance of 8 inches is highly recommended. The PDS carrier should not be mounted to or suspended from the ceiling wherever possible.
- j. The PDS carrier is not permitted to be mounted to, be fastened to, or be supported by suspended ceiling (i.e., acoustical tile ceiling) hardware.
- k. When the PDS must cross a Black wire line or AC power line, it shall be at a right angle and shall maintain a minimum separation of ½ of an inch (1.25 Centimeters [cm]).
- l. Route the PDS carrier so the installation does not traverse across or block access to any window, door, air conditioning duct, lighting fixtures, fire suppression equipment, or utility openings in the walls, floor, or ceiling of any room or hallway.
- m. Do not route or install the PDS in or through public areas where personnel could linger without scrutiny (i.e., break areas) or that have an expectation of privacy (i.e., restrooms).
- n. Do not route or install the PDS through areas (janitor closet, mechanical room, electrical room, etc.) where access limitations would preclude the ability to conduct daily visual inspections by the building occupant, unless there is no other option.
- o. Maintain a separation of 2 inches (5 cm) between Red and Black wire cable for parallel runs less than 100 feet (30 m). Increase this separation to 6 inches (15 cm) when the parallel run is greater than 100 feet (30 m).³⁸

3.4.7 Marking of PDS

The PDS shall have no external markings that reveal the classification level of the information transmitted through the PDS.³⁹ The PDS shall not be painted, covered with wall paper, or covered with any other material that will obscure or hide signs of penetration, tampering, or any other anomaly causing the deterioration of protection safeguards. Use of secure raceway (i.e., Holocom, Wiremold) that is painted at the time of manufacture is acceptable.

3.4.8 Boxes (UDB, Pull, Junction)

- a. Drop, pull, junction, and any other type of box or housing containing pre-punched knockouts are not acceptable for use in a PDS.⁴⁰
- b. UDBs shall be constructed of 16-gauge steel or greater, and corner seams shall be welded. UDB dimensions shall have a minimum dimension of 6 inches wide by 6 inches high by 4 inches deep or greater. All pull, drop, and junction boxes shall be the proper size

³⁸ NSTISSAM 2-95, Section 3.

³⁹ AR 380-5, Chapter 7, Paragraph 7-8.a, Page 81.

⁴⁰ NSTISSI 7003, Annex B, Paragraph 4.a(1)(c).

SIPRNET Technical Implementation Criteria

September 2013

to accommodate the number and size of PDS carrier connections and cables planned for placement in/through the box.

c. The UDB shall be equipped with a built-in locking hasp to secure the access door in the closed position, using a changeable combination padlock built to Federal Specification FF-P-2740A with Amendment 1. The Sargent & Greenleaf, model 8077 padlock is the only lock currently meeting this specification.⁴¹ Holocom pull and junction boxes, equipped with properly installed internal locking mechanisms to secure the access door, are not required to have locking hasps.

d. UDBs shall be mounted between 36 inches and 66 inches above the finished floor, measured to the bottom of the UDB. If modular furniture and/or partitions are used in the room, mount the UDB above the level of the furniture and partitions. The UDB shall be visible at all times after the furniture and/or partitions are installed.

e. UDBs shall be capable of having a standard, single or double gang, telecommunications cover plate installed inside the UDB for termination of the cables in standard access jacks. The cover plate shall be sufficiently recessed, so it does not interfere with the proper closing and locking of the access door when no user cables are connected to the front of the network access jacks in the cover plate.

f. Pull and junction boxes shall be constructed from 16-gauge steel or greater and have either a lockable door with a built-in hasp (same as UDB) and internal hinges or a removable cover secured with screws. Epoxy shall be applied on all edges of removable cover and screw tops. Mount pull and junction boxes flush to the wall partition surface.

3.4.9 Epoxy

Use a two-part, self-hardening epoxy equivalent to the 3M DP-420 series, with at least a 20-minute working life on all joints and seams. The epoxy color shall contrast with the PDS finish. Clear (light yellow color when cured) epoxy shall be the default for secure raceway (Holocom or Wiremold) PDS and gray or black for conduit carrier. Local security personnel may specify the color of epoxy. Epoxy that remains completely clear upon hardening shall not be used.

Apply one bead of epoxy to each end of the compression and box connector fittings connected to EMT. Apply three horizontal beads applied between the two vertical beads at the 10, 2, and 6 o'clock position (60, 180, and 300 degrees) on EMT compression fittings.

Apply one bead of epoxy completely around the surface where two components meet or connect to each other. Figure 1 shows examples of epoxy beads on connectors.

⁴¹ E-mail, GSA, Mr. Pollock, 4 May 2010.

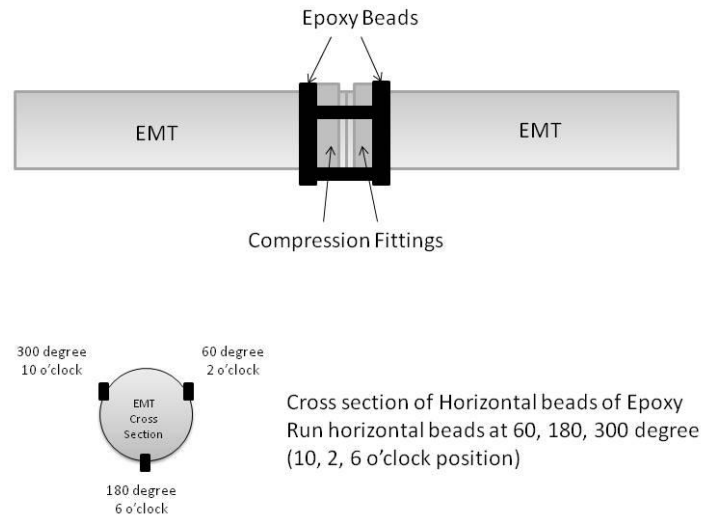


Figure 1. Epoxy Beads on Connectors

3.5 SIPRNet Architecture and Design

3.5.1 General

There are three basic physical architectures for the distribution of SIPRNET inside a facility or EUB: [distributed PDS](#), [SIPRNET café](#), and [individual COMSEC](#). Each alternative offers various advantages and disadvantages in physical security, ease of operation, and cost.

Regardless of the physical architecture chosen, all implementations of SIPRNET must take into consideration the requirement to evaluate and control compromising electromagnetic emanations, commonly referred to as TEMPEST.⁴² The required time to address TEMPEST issues is determined during the design phase of the SIPRNET system.

Of particular concern for TEMPEST issues is the size of the inspectable space around SIPRNET assets and the type and location of all fixed Radio Frequency (RF) transmitters and antennas within 10 m of classified facilities.⁴³ While additional information is presented in [Section 3.9.2\(c\)](#) of this document, a TEMPEST Countermeasure Review (TCR) and the PDS technical review(s) by the USACTTA are required as part of the Certification and Accreditation (C&A) package for classified networks.

⁴² AR 380-27, Control of Compromising Emanations, 19 May 2010.

⁴³ AR 380-27, Chapter 4, Paragraph 4-1.

3.5.2 Design Considerations

SIPRNET is a MC system. As such, the reliability and availability of the user's access to it must always be considered when designing and installing a SIPRNET system. The importance of reliability and availability is greater than is found on a non-MC system, such as Unclassified but Sensitive Internet Protocol Router Network (NIPRNET).

Physical security must be addressed during the design and installation of all SIPRNET access points to ensure the system can successfully pass the physical security inspection required for accreditation.

In all locations where the equipment used for SIPRNET access provides service to multiple users, such as the installation POP or a building POP, the equipment shall be provided with AC power from a dedicated circuit breaker. If the equipment is contained in multiple equipment racks, cabinets, or other types of security housing, each rack shall be provided with AC power from a dedicated circuit breaker.⁴⁴ For SIPRNET equipment with a redundancy capability, provide AC power from dedicated circuit breakers from separate technical power sources.

Provide backup power (generator or an Uninterruptible Power Supply [UPS]) to buildings that service multiple users, such as the installation POP or an MC building POP. Size the UPS to provide 2-8 hours of backup power for all SIPRNET equipment to meet Assured Services Local Area Network (ASLAN) power requirements.⁴⁵

3.5.3 Power over Ethernet (POE)

Include POE for Voice over Secure Internet Protocol (VoSIP) when designing a SIPRNET system (see [Section 3.10](#)) for each building POP. All network switches will be POE switches. Size the POE switch for 75-percent usage.

A POE switch consumes more power than a regular network switch, and this will impact the UPS sizing and HVAC cooling requirements. The UPS may require resizing upwards to maintain the required run time during a power outage. The additional power consumed by the POE switch also generates more heat (approximately 2,800 British Thermal Unit (BTU)/hour per 48-port POE switch, depending on the switch type). In a closed environment, such as inside a closed equipment cabinet or IPS container, additional cooling capacity may be required.

3.5.4 Distributed PDS

A distributed PDS is one where the PDS carrier routes from the source (the distribution switch housed within a secure TR or IPS container) to the destination (UDB locations) in a branching fashion. The top portion of [Figure A-7](#) depicts an example of a distributed PDS architecture consisting of 11 users with SIPRNET access. There are 11 UDBs, each containing one SIPRNET connection. In a room with multiple users, or where a network printer or other devices are required, a UDB may contain several user connections.

⁴⁴ Technical Criteria for the Installation Information Infrastructure Architecture, Paragraph 2.5.1.5.

⁴⁵ UC Framework 2013, Section 7.8.4

A vast majority of facilities do not meet CAA standards and will require a hardened carrier PDS to implement SIPRNET using the distributed architecture. Due to the large amount of PDS material required in the distributed architecture, and the labor required to install it, the overall cost of the installation in non-CAA facilities is relatively high.

On the operational side, users may access SIPRNET from their desk when SIPRNET access is required. This does not address the issues of where the classified laptops or hard drives will be stored when not in use. It is impractical for users to have their own storage safe for their computer hardware and any classified documents or material they may have. Typically, there will be one central location containing the storage safe(s) that users will have to access before and after using SIPRNET. Only the actual usage time occurs at their desk.

From the operational security aspect, the security zone for the building is all of the areas where SIPRNET is used, or transits through, via the PDS is located in the distributed architecture. In a widely distributed architecture, the security zone could encompass almost the entire building. There may be several different types of access areas the PDS transits through within the building security zone. Within the overall security zone, the PDS must be monitored and inspected according to the type of access areas that it traverses through. At each user drop location, care must be taken to ensure computer screens are properly oriented and shielded to prevent the unintentional displaying of classified material. Personnel must be aware of, and follow, the security procedures developed within this large security zone. As the SIPRNET architecture distributes throughout the building and the security zone increases in size, the potential for an inadvertent security violation increases.

In a large distributed PDS using Category (CAT) 6 cabling, the effective cabling distance for Ethernet (90 m) may be exceeded. To ensure all user access connections are within the Ethernet specifications, there are two solutions: fiber optic cabling or intermediate switches.

3.5.4.1 Fiber Optic Cabling

One solution is to change the cabling for the distant user drops from copper to multimode or single mode fiber optic cabling, which can support much longer Ethernet distances (2 Kilometers [km]). This requires different port and connector types (fiber) on the building POP switch, as well as in the user computers. Another solution is the use of intermediate, or workgroup, switches along the PDS route to extend the effective Ethernet distance. These switches act as repeaters and multiplexers by regenerating the Ethernet signal, giving another 100 m of connectivity, and they extend the connectivity of one signal cable from the building POP to multiple distant users.

3.5.4.2 Intermediate Switches

When using intermediate switches, the switches must be afforded the same or a greater level of physical protection as the SIPRNET cabling inside the PDS. In some cases, the intermediate switches may be placed inside a large, wall-mounted, secure equipment enclosure (e.g., Holocom SEC-WM-04U-H1) or a small, 8-port switch in a pull box somewhere along the PDS carrier route. In other cases, an IPS container must be installed, and the PDS carrier must be routed to/from it. The use of the IPS container is more expensive than the large PDS box, and it requires additional floor space in an office or TR. The use of the large, secure equipment enclosure is less expensive, and it can be wall-mounted along the PDS carrier route, making it more convenient to use. In this case, the

SIPRNET Technical Implementation Criteria

September 2013

secure equipment enclosure shall meet the same standards as the UDB, including being secured with the same type of changeable combination padlock. The use of the secure equipment enclosure to house the intermediate switch must be closely coordinated with the local security personnel, as it will require a risk acceptance from the DAA to obtain accreditation for the entire system. The PDS design must also be staffed through the USACTTA for technical review (see [Section 4.2.2](#)) prior to installation or procurement of materials.

3.5.5 Secure TR

The main secure TR is where the building POP is located. All distribution to users within a building will originate from either the main secure TR or secondary secure TR.

One main secure TR is required for each building, and wherever possible it must be located adjacent to the main NIPRNET TR. Whenever possible, locate each secondary secure TR adjacent to a NIPRNET TR. Secure TRs should be stacked directly above one another whenever possible in multifloor buildings to create one extended security perimeter.

The interior dimensions for the secure TR will be a minimum of 6'6" by 8'6" and will be constructed IAW secure room, open storage area standards (see [Section 3.3.5](#)). This will provide a 30-inch clearance between the front and back of the equipment rack (24 inches wide by 36 inches deep) and 24 inches on each side of the rack. Increase the square footage to 3' by 8'6" for each additional equipment rack or cabinet that is installed in the secure TR. The secure TR ceiling shall be exposed to the structure above, unless a finished Gypsum Wallboard (GWB) ceiling is used. The minimum height of a finished ceiling shall be 10 feet above the finished floor level.

The door shall be constructed of solid wood core or hollow metal, a minimum of 36 inches wide by 80 inches tall, and 1 3/4 inches thick. The door shall also be constructed without a doorsill, louvers, or windows; shall be hinged to open (up to 180 degrees) outward with NRP or welded/pinned pins; and shall have an auto closer. The lockset shall consist of a built-in, GSA-approved, changeable combination lock that meets Federal Specification FF-L-2740A (e.g., Kaba Mas CDX-09 or equivalent).

Equipment racks shall be floor-mounted, 84 inches (2,100 Millimeters [mm]) high, 19 inches (475 mm) wide, and located at or near the center of the TR. Equipment racks are normally 7 feet (2.1 m) high but may be adjusted for special conditions. If mounting requirements for oversized equipment are anticipated, 23 inches (580 mm) may be substituted. In narrow or crowded secure TRs, equipment racks may be floor-mounted and adjacent to a wall, but shall provide a minimum of 36 inches (900 mm) of space in front of and behind the rack, as well as behind any installed equipment. A minimum side clearance of 24 inches (600 mm) shall be provided on end racks.

The equipment cabinets shall be located in a room that permits front and rear doors to be opened a minimum of 90 degrees. To prevent casual access to the equipment, the doors must be lockable. The use of standard cabinet locks and hardware is permitted. Removable side panels on the cabinet may be used to supplement, but not replace, the accessibility provided by the front and rear doors. Equipment cabinets shall have ventilation ports on the sides (at top and bottom) and/or the front and rear. In addition, exhaust fans shall be installed to draw hot air out of the cabinet from the top through the top ventilation ports. The top vents, whether in the top of the cabinet or the top of the cabinet sides, shall contain

the exhaust fans. Top vents that allow air to enter the top of the rack and immediately exhaust out of the top of the rack through the fans are not permitted. The fans shall be the proper size to provide sufficient airflow, based on the heat load of the equipment and the air temperature in the surrounding room, to keep the air temperature inside the cabinet no more than 15 degrees Fahrenheit (F) above the room temperature.

Note: Equipment racks are normally located in secure TRs, where as equipment cabinets are used in server-type rooms or in rooms that have multiple classified domains collocated in the same secure room.

Connection from the secure TR to the main NIPRNET TR will occur via a single, 2-inch, trade-size, steel or PVC conduit.

Do not route utility piping or ductwork through the Secure TR space, which does not provide utility service to the secure TR. Doing so may require hardening of the utility lines, which will degrade service (e.g., each hardening of an HVAC duct reduces the airflow by approximately 20 percent).

Utility openings, such as ducts and vents, will be kept at less than a person-wide opening (96 square inches). Openings that exceed 96 square inches will be hardened IAW Military Handbook (MIL-HDBK) 1013/1A. Openings that exceed 96 square inches with one dimension less than or equal to 6 inches do not require hardening.

A communications signal, Telecommunication Grounding Busbar (TGB) must connect to the main TR signal, Telecommunications Main Ground Busbar (TMGB) via properly sized ground wire (see MIL-HDBK-419-A). Daisy chaining multiple TGB bonds in series to the TMGB is not acceptable, and home runs are required.

A minimum of one dedicated, 120-Volt Alternating Current (VAC), 20-Ampere (amp) or 208-VAC, 15-amp circuit breaker is required per SIPRNET equipment rack in addition to convenience outlets within the secure TR. Add an additional dedicated circuit breaker from a different power panel if power redundancy is required. Convenience outlets within the secure TR shall not share the same power circuit as convenience outlets exterior to the secure TR.

The secure TR ceiling shall be exposed to the structure above or finished with GWB. An acoustical tile ceiling is not acceptable.

Light fixtures shall be mounted a minimum of 9 feet (3 m) above the finished floor and shall provide a minimum of 50 foot candles (538 Lumens [lux] of illumination measured 3 feet (1 m) above the finished floor. Recessed lighting is not acceptable.

Fire-rated, plywood backer boards are not required on any wall partition. However, the fire rating stamp must remain visible if fire-rated plywood is installed on any wall partition.

Provide 24/7/365 HVAC environmental controls to provide positive air pressure within the TR, maintain a constant temperature not to exceed 85 degrees F, and maintain relative humidity between 50 and 70 percent.

3.5.6 Building POP

The building POP is the terminology used to identify the location where the main network switch and encryption (COMSEC) device is located in the main secure TR, a SIPRNET café, or IPS container. SIPRNET distribution throughout a facility originates from the building POP.

SIPRNET Technical Implementation Criteria

September 2013

The network equipment installed for the building POP varies widely, depending upon the actual site conditions and requirements. The simplest configuration consists of a network encryption device (COMSEC) and a distribution switch. A more complicated configuration may consist of a router, the firewall(s), a Network Intrusion Detection System (NIDS), a Network Intrusion Prevention System (NIPS), and a distribution switch.

Depending on the criticality of the building, an UPS may be required to ensure power is maintained to the building POP should power fail. An alternative is to have the building on a backup power source.

3.5.7 Building POP Physical Security

The security housing is the barrier that prevents unauthorized personnel from accessing the POP equipment. It may be an equipment cabinet; a GSA-approved IPS container; a dedicated, secure TR; or even an entire building.

There are several key factors that must be considered in the selection of the POP security housing, which is discussed in following sections.

In an open storage area at the Secret (or higher) level, classified material may be stored openly (i.e., not in a GSA-approved safe), as the room itself is the security container. In this case, the network equipment may be placed inside a standard equipment cabinet. The equipment cabinet shall be four-sided with a top, and each access door shall be equipped with keyed locks to prevent casual access to the network equipment. An IPS container may be used if additional security is desired.

3.5.8 SIPRNET Café

The use of a SIPRNET café is ideal for a group of users that only require occasional SIPRNET access. The bottom portion of [Figure A-7](#) depicts an example of a SIPRNET café consisting of three users with SIPRNET access. In this type of layout, all the equipment and PDS are located in one room, the café. The storage safe(s) for the user terminals and other classified material are also typically located in the SIPRNET café. The type of PDS carrier used depends upon whether the café is an LCA, a CAA, or an open storage area. A SIPRNET café shall not be placed in a [UAA](#). The café layout takes advantage of the fact that although there may be multiple SIPRNET users in a facility, not all of them are accessing SIPRNET at the same time. The SIPRNET access is a shared asset. There are fewer UDBs required, but each one typically contains two network connections.

The use of the café concept provides SIPRNET access capability to authorized, occasional users within the entire facility and reduces the size and cost of the PDS infrastructure. This reduces the associated risk by reducing the necessary physical security upgrades to the facility and by consolidating the local security procedures to a smaller area.

The café room selected must be sized to accommodate the [IPS container](#) (see Paragraph 4.7.4[b][2]) used to house the network electronics, the terminal equipment (laptops, Hard Disk Drive [HDD], etc.), storage safe(s), and the maximum number of users expected to be using the café at any given time. In this example, a six-user café has been selected. Thus, the UDBs located around the room provide sufficient spacing to allow for six users to sit at desks or tables. Typically, a UDB is placed in the middle between each pair of desks/tables. The spacing between users must allow sufficient space to work and provide a “buffer zone” between users to prevent inadvertent viewing of another user’s work (thus maintaining the “need to know” aspect of access to classified information). Each user shall have an absolute

minimum 3-foot-wide by 2-foot-deep working space allocated (the size of a folding field table), although a 4- to 6-foot-wide space is highly recommended.

From an operational aspect, to use the SIPRNET, users must leave their offices and go to the café. This is not too different from the distributed layout in which users must leave their offices to obtain their classified computer hardware from the storage safe(s). The difference is that users must remain in the café, away from their desks, during SIPRNET usage.

As all of the PDS and safes are in one location, the security zone that must be monitored and checked is much smaller, making it relatively easy to turn the entire area into an [LCA](#) or a [CAA](#). As café rooms are often in areas of the facility that are not under constant observation during the duty day, the chosen café room should be a [CAA](#). With all of the classified equipment and material in one room, there is typically less chance for an inadvertent security violation.

3.5.9 Individual COMSEC

In this physical architecture, SIPRNET users are provided with their own COMSEC device. This eliminates the need for a PDS as the signal is encrypted (i.e., Black) until it reaches the user's COMSEC device.

Measures must be taken to prevent the inadvertent viewing of classified material. While accessing the SIPRNET the following requirements must comply (local requirements may vary):

- a. The user's computer screen must be oriented, so it cannot be viewed from outside the office or from the office door.
- b. The windows in the room must be covered while SIPRNET is in use in the office to prevent inadvertent viewing of classified material from outside the room.
- c. If personnel other than the user work in the office, they should possess at least a Secret-level security clearance.⁴⁶
- d. Laptop computers and the HDDs from desktop SIPRNET computers must be stored in a GSA-approved safe when not in use. The COMSEC device must also be stored in the safe or be rendered unclassified. Rendering a device unclassified is normally accomplished by removing the Crypto Ignition Key (CIK), a small round electronic key about 1 inch in diameter and 1/2 inch thick, but the actual process is device dependent. As the COMSEC device remains a Controlled Cryptographic Item (CCI) even when unclassified, it must be properly secured. It must be secured in a GSA-approved safe or stored in a locked container inside a locked room. The CIK is also unclassified when removed from the COMSEC device and must be secured in a similar fashion.

There are a number of different encryption devices that can be used as individual COMSEC devices, each with different capabilities and unit cost. More details on the capabilities are provided in [Section 3.8](#) and [Appendix B](#).

⁴⁶ AR 380-5, Section 7-20, Paragraph d(1).

SIPRNET Technical Implementation Criteria

September 2013

More than one person may use an individual encryptor. A separate profile is required on the SIPRNET workstation for each user.

An individual encryptor may not be the most cost effective solution when there are numerous SIPRNET users not widely spaced. Where there are few users that are spaced far from each other, it is probably more cost effective to use an individual encryptor for each user. Number and location of users and the size of building are factors to consider when selecting the type of architecture.

3.6 Tunneling

Tunneling is a means of connecting two remote network segments together by using a third network as a means of transport without the transport network being able to read the packets. It is the preferred method of extending SIPRNET access from one building (typically the post SIPRNET POP) to another (typically a remote tenant facility on post) across the unclassified post CAN. To maintain the confidentiality of the classified information on the SIPRNET, the traffic is encrypted using INE before transport across the CAN.

When tunneling across the CAN, it is incumbent upon the local communications provider, such as the NEC, to ensure the transport network (the CAN) carrying the SIPRNET traffic maintains the highest network reliability and availability as possible to reduce potential outages for the MC traffic on the SIPRNET. For Army posts that have already completed the I3MP process and are equipped with redundant links and equipment, this does not pose a problem. For posts that have yet to complete the I3MP upgrade process, a separate means of transport may be required.

On an IP-based network, such as the post CAN, the IP packets of data are occasionally discarded, resulting in either the loss of the data or the need to retransmit it. Encrypted SIPRNET traffic should be placed in a separate VLAN from other network traffic. Although it physically traverses the same cables as the rest of the CAN traffic, the SIPRNET traffic is logically separated from the CAN traffic by the network routers and switches. The SIPRNET VLAN traffic should also be given priority over other traffic at common processing locations, such as routers and switches, to further reduce the possibility of dropped packets. In addition, if Multiprotocol Label Switching (MPLS) is used, the SIPRNET traffic should be put into an MPLS VPN with bandwidth guarantees to provide further protection.

To use VLANs on the CAN infrastructure, each router and switch in the CAN should be configured for VLANs. The ports on the INE connect to the EUB switch, which is configured for VLAN access, while the intermediate equipment is configured for VLAN trunking. Only equipment connected to a SIPRNET VLAN port, such as the INE in various buildings, will receive the IP packets sent on the VLAN.

3.7 IPS Container

Building POP network equipment shall be housed in an IPS container in an area that is not accredited for open storage. The IPS container is a GSA-approved, Class 5 safe that is designed to securely house classified network equipment and has predrilled penetrations for signal and power cabling, along with ventilation and cooling openings. Companies that manufacture standard classified document storage safes typically manufacture IPS containers.

Classified material and offline, classified computer hardware (laptops and removable hard drives) are not permitted to be stored inside an IPS container. These items must be stored in a separate GSA-approved, Class 5 or Class 6 safe.⁴⁷

3.7.1 Floor Loading

When using an IPS container, floor loading shall be considered. For a POP that is in an IPS container, it can be in the 1,000- to 2,000-Pound (lb) range. To ascertain whether floor loading will be an issue, determine the total weight of the POP and security housing footprint, and give that information to the building structural engineer for analysis. For floor loadings exceeding 250 Pounds Per Square Foot (lbs/sq ft), a metal spreader plate under the IPS container may be required to distribute the floor load to acceptable levels. The appropriate structural engineering authority shall make all such determinations.

3.7.2 Room Accessibility

An IPS container is a large bulky item that must pass through the building's doorways, stairways, and hallways along the delivery path. Depending upon the room's location, special equipment may be required to deliver the IPS container. The size and weight of the IPS container must be considered when selecting a room and the path to the room.

The room selected for the IPS container shall have a minimum 3-foot door entrance. All hallways along the delivery route should be at least 5 feet wide. All other doorways along the delivery route should be larger than the room entrance if possible, but never smaller.

3.7.3 Rack Space

Sufficient rack space for all equipment must be available inside the security housing. Standard rack spaces are available in widths of 19 and 23 inches, with 19 inches being the most common in network equipment. Rack space normally is expressed in terms of Rack Units (RUs). The height of the IPS container shall provide sufficient space for all equipment, patch panels, and cable management equipment, as well as provide space to access the equipment.

3.7.4 Equipment Accessibility

The IPS container shall be deep enough (front to rear) to accommodate the longest piece of equipment with space for cable connectors and the proper bending radius on the front and rear of the equipment. When installed in the security housing, the network equipment must be accessible for initial cabling and connection, as well as for any future maintenance or replacement actions. The equipment rack inside an IPS container is either fixed or movable.

Fixed racks are fastened to the sides on the inside of the IPS container. Once the equipment is mounted in the rack, the connections on the rear of the equipment are only accessible by reaching through the front of the rack and blindly feeling around. When fixed racks are used, extra rack space (RUs) must be allocated to allow for this. At least 3 RUs shall be available below or above each 4 RUs of equipment. These cabling spaces shall remain empty of equipment or blank panels to allow accessibility and to promote air circulation.

⁴⁷ CNSSI 4005 Section VII.D, Paragraph 6, 22 August 2011.

SIPRNET Technical Implementation Criteria

September 2013

Movable racks are those where the entire rack assembly slides or rolls out from inside the IPS container on some sort of track mechanism. This allows access to all sides of the equipment. When this type of rack is used, the movable rack shall extend far enough out from the front of the IPS container to provide a minimum of 6 inches of clearance from the rear of the equipment to the front of the IPS container. A cable control system shall prevent cables draping from the back of the movable rack to the back of the IPS container (e.g., AC power and user drop cables) from falling to the bottom of the IPS container and being pinched or damaged by the rack as it retracts inside the container. This cable control system shall be implemented without causing undue stress or stretching of the cables. [Appendix G](#) provides the details of some IPS containers with movable racks.

3.7.5 Ventilation

The heat generated by the network equipment inside the security housing may cause the interior air temperature to rise to dangerous levels, causing the network equipment to fail if the ventilation does not dissipate the heat. Active cooling measures (i.e., interior air conditioning) are not permitted due to the potential for equipment failure. Passive heat exchangers are permitted in combination with fans. The room or space where the security housing is located shall have sufficient cooling and ventilation to maintain an ambient temperature of 75 degrees F or lower.

An IPS container shall be provided with a similar ventilation system. The fans shall be the proper size to provide sufficient airflow, based on the heat load of the equipment and the air temperature in the surrounding room, to keep the air temperature inside the cabinet no more than 15 degrees F above the room temperature. The exhaust fan shall exhaust air from the upper portion of the IPS container. The air intake may be located on the bottom of the IPS container, the side opposite the exhaust vent, or the bottom of the same side as the exhaust vent.

The IPS container must be located in the room to allow sufficient clearance around all ventilation ports for proper airflow. At least 4 inches of clearance shall be maintained between all vents and any obstructions, such as walls, filing cabinets, storage boxes, other cabinets, etc.

The top 1 RU of rack space in the IPS must remain empty and may not contain passive equipment, such as patch panels, and 1RU of rack space must remain empty above any active equipment to help prevent overheating of equipment. The bottom of any rack shelves used inside the IPS shall be vented rather than solid, so airflow is unrestricted.

All air intakes shall be filtered to prevent dust and dirt from being drawn into the security housing. The filter shall be easily removable for cleaning and/or replacement.

3.7.6 Noise Suppression

The noise generated by the ventilation system must be accounted for, especially in a working office environment. When the security housing is located in the same area where personnel will be working, such as in a SIPRNET café, the fan and equipment noise from the security housing shall be attenuated down to the level of ordinary conversation 60 Decibels (dB) or less. When located in a TR or other unattended area, noise suppression is optional.

3.7.7 Exterior Cabling Entrance

The network equipment requires three types of cabling that exit/enter the security housing, AC power, incoming signal to the Black side of the COMSEC device, and the Red-side user drop cables for network access.

The security housing shall have two separate openings for cable entrances, one for Black cables (AC power and incoming Black signal) and one for Red cables (user drop cables to the UDBs).

For an equipment cabinet used as the security housing, the Black cables shall be protected where they enter the cabinet to prevent cable damage from the edges of the cabinet. The carrier for the Red cables shall be continuous from the UDBs into the equipment cabinet to shield the cables from view. The carrier shall enter directly into the cabinet, or the proper type of fitting be used, to provide the same level of protection to the Red cables.

For an IPS container used as the security housing, the Red cable entrance shall be equipped with a heavy-duty, lockable box covering it to allow the termination of the PDS duct without exposing any of the Red cabling as it routes from the PDS carrier into the cable entrance box. The Red cable's entrance box shall meet the same physical security requirements as the UDBs.

3.8 Encryption Devices

3.8.1 General

Encryption devices (COMSEC) scramble the local classified (Red) data using an electronic key that must be loaded into the device, along with a software algorithm hard-coded into the device, to produce an outgoing unclassified (Black) data stream. They also operate in the reverse direction to decrypt the incoming Black data stream. The Black data stream may be safely transmitted over a communications means without fear of compromise. Even if it is intercepted, the Black data stream will not reveal classified information.

Within the U.S. Army, all COMSEC devices must be obtained through CSLA.⁴⁸ CSLA has established the Information Systems Security Program (ISSP) for this purpose. Orders for COMSEC devices must be validated through the ISSP Website.

COMSEC device(s) must be on the requesting organization's Modified Table of Organization and Equipment (MTOE)/Table of Distribution and Allowances (TDA) before CSLA will process the requisition. Submit the completed DD Form 4610-R, *Equipment Change in MTOE/TDA*, with the requisition package to CSLA for new COMSEC requirements.

Note: The ISSP Website is at <https://issp.army.mil> (requires Army Knowledge Online [AKO] login to ISSP Website).

⁴⁸ AR 25-2, Chapter 6, Paragraph 6-1.a(4), Page 52.

SIPRNET Technical Implementation Criteria

September 2013

3.8.2 Types of Encryption Devices

A wide variety of encryption devices designed for use with different transmission methods exist. Generally, devices used to encrypt and decrypt information are referred to as COMSEC or encryption devices.

Note: When used to encrypt NSI, the COMSEC devices must be certified by NSA for Type 1 encryption.⁴⁹

There are three basic groups or types of encryption devices, which are discussed in the following sections. Some COMSEC devices are capable of operating in more than one of these configurations.

3.8.3 Link Encryptor

Link encryptor is a family of encryption devices designed for use on link and trunk circuits typically found in long-haul transmission. They are usually designed for serial circuits, such as Transmission Carrier 1s (T1s), Digital Signal Level 3s (DS3s), or other high capacity circuits. As these high capacity circuits are often the aggregate of many smaller circuits, they are also referred to as trunk encryptors.

Note: An example of a link encryptor is the KIV-7M.

3.8.4 INEs

Encryption devices designed for use in IP-based data networks are referred generically to as INEs. Although INEs have been used since the early 1990s, it is only since 2000 that their use has become widespread. This has resulted in the need for interoperability and IA standards.

Note: Examples of in-line link encryptors are the KG-175 Tactical Local Area Network Encryptor (TACLANE) family, KG-240 ViaSat, and KG-250 RedEagle family.

3.8.5 Individual Mobile Encryptors (IMEs)

An IME is a network encryptor designed for use by a single user rather than a network of users. IMEs are particularly useful for personnel who must travel (i.e., are mobile) and need access to SIPRNET while traveling. However, when used in this fashion, the physical security of the classified computer hardware and the IME often present sufficient disadvantages to outweigh the advantage of secure mobile computing. They are also useful for the individual COMSEC architecture described in [Section 3.5.5](#).

Note: The KOV-26 (Talon) and KIV-54 (SecNet-54) are examples of IMEs.

3.9 Secure Wireless Local Area Networks (SWLANs)

3.9.1 General

WLANs use a radio link to transmit and receive information between two wired LANs (referred to as Bridge mode) or between a wired LAN and an end user (referred to as Access mode). The standard used for wireless networking is the Institute of Electrical and Electronics Engineers (IEEE) 802.11 series of standards. This document is concerned with

⁴⁹ AR 25-2, Chapter 6, Paragraph 6-1.a, Page 52.

WLANs (802.11a/b/g/n) and not with other wireless technologies, such as wireless personal area networking (802.15, or Bluetooth) and Worldwide Interoperability for Microwave Access (WiMAX) (802.16), pagers, cellular telephones, etc.

The DOD's policy is that unclassified Wireless Local Area Network (WLAN) systems must be standards-based and IEEE 802.11 compliant. In addition, to help protect the information in the transmitted radio signal, the system must use Wireless Protected Access 2 (WPA2)-enabled devices that implement the Advanced Encryption Standard (AES).⁵⁰ This helps protect the WLAN from unauthorized access at the link layer.

NSA must certify the equipment used in classified SWLANs, and the SWLAN must use NSA Type 1 encryption devices, as well as be protected by physical security measures appropriate for the classification of the NSI being processed.⁵¹

Contact the supporting USACTTA prior to installing Wireless Access Points (WAPs) in a classified processing area.

3.9.2 Wireless TEMPEST Issues

As the transmission link is a radio signal, any receiver, whether authorized or not, that is within range of the signal can capture it. The radio signals penetrate most building materials, so the unauthorized receiver does not necessarily need to be in the same facility. TEMPEST is the name given to the evaluation and control of the compromising emanations from telecommunications and automated IS equipment. To prevent unauthorized persons from receiving, reading, and exploiting the information transmitted, special security measures must be taken when implementing WLANs. These special measures are generally known as TEMPEST countermeasures, an unclassified codename referring to the measures taken to contain the compromising electromagnetic emanations within the controlled inspectable space around the equipment or facility.

Due to multiple technical security issues, the use of wireless communications devices, such as two-way radios, pagers, Bluetooth devices, etc., is prohibited in areas where classified information is discussed or processed, such as SIPRNET rooms.⁵² These devices may be permitted if certain conditions are met⁵³ and are approved by the DAA after a technical review by the USACTTA. Additional and more detailed information regarding TEMPEST issues may be obtained from the USACTTA.⁵⁴

The location of any fixed RF transmitting device or antenna within 10 m of a facility with fixed Red processing equipment is of special concern in the area of TEMPEST.⁵⁵ A fixed RF device or antenna is one that remains in a specific location and orientation for an extended period. A laptop computer that is placed on a desk at the beginning of each day

⁵⁰ DODI 8420.01, Enclosure 3, Paragraph 1.a.

⁵¹ DODI 8420.01, Enclosure 3, Paragraphs 4.a and 4.b.

⁵² AR 25-2, Paragraph 4-29.a, Page 46.

⁵³ Unsigned Memorandum, IAMG-CIC-OP-USACTTA, Paragraph 3.

⁵⁴ USACTTA TEMPEST Information Handout.

⁵⁵ AR 380-27, Chapter 4, Paragraph 4-1.

SIPRNET Technical Implementation Criteria

September 2013

and removed at the end of each day is not considered fixed. If the laptop is placed on the desk and not moved each day, or is placed in a docking station that is left in the same position on the desk every day, it is considered to be fixed.

Contact the USACTTA for specific installation guidance for any transmitters in or near classified areas.

3.9.3 SWLAN Design Considerations

A SWLAN provides a SIPRNET access solution that may reduce the time and manpower required to setup and operate a secure communication infrastructure. Although wireless solutions may have merit in certain situations and applications, wireless SIPRNET may not be the best choice for the normal SIPRNET user. The minimum separation distance limitations as set forth in National and Army policy⁵⁶ may limit the number of secure wireless devices that can be installed in an area.

The SWLAN is used to provide wireless SIPRNET access that is extended from the wired portion of the local SIPRNET infrastructure. The connection of the SWLAN to the wired portion of the local SIPRNET must be part of the SWLAN design.

As shown in [Figure A-8](#), three basic architectures are approved for use in SWLANs.⁵⁷ An actual SWLAN may also be a combination of the different architectures.

The wireless access architecture is the one most common. This is when the Wireless Access Point (WAP) operates in Client Access mode, allowing users to connect to it via the wireless connection while the wired side of the WAP connects to the network. It functions to grant wireless users access to a wired network.

The wireless bridging architecture is when two or more WAPs operate in Bridge mode. This architecture functions to extend a wired network from one location to another via the wireless connection between the WAPs. Operated in this mode, the WAPs normally will not allow wireless clients to connect to them.

The peer-to-peer architecture is an “ad-hoc” network setup between local clients via its wireless capabilities. It does not require a WAP, as all connections are between the clients. This type of wireless architecture is not recommended, as by directly linking the clients, all network security and monitoring devices are bypassed, allowing a virus or Trojan Horse program on one client to easily spread to the others.

While setting up a SWLAN appears simple (from a wireless client perspective it is) there are a number of management burdens associated with SWLANs that increase the Operations and Maintenance (O&M) burden of operating a SWLAN. Each requires the expenditure of equipment or manpower resources beyond the actual wireless client connection.

A Wireless Intrusion Detection System (WIDS) must be employed throughout the entire area of operations (not just the area where authorized WLANs are used) to monitor for unauthorized WLAN usage, whether secure or non-secure.⁵⁸ WIDS are wireless devices and

⁵⁶ NSTISSAM 2-95, NSTISSAM 2-95A, and Memorandum, IAMG-CIC-OP-USACTTA.

⁵⁷ DISA Wireless STIG.

⁵⁸ DISA Wireless STIG.

must be reviewed by the USACTTA prior to installation when operating in presence of classified processors. The WIDS must monitor all WLAN frequency bands available, not just those used locally for wireless access or bridging. For a large area, such as the metropolitan area size of most cantonment areas on Army posts, this can represent a large number of devices.

A WIDS server must be employed to continuously monitor the deployed WIDS.⁵⁹ This requires personnel to periodically review the server logs for signs of unauthorized WLAN activity and take the appropriate action.

Only authorized wireless systems may be employed in a WLAN.⁶⁰ For a SWLAN, only devices with NSA Type 1 encryption may be used.⁶¹ The only device currently authorized for use in a SWLAN is the SecNet 54 (KIV-54) and the Talon (KOV-26).⁶² The SecNet 11 Plus is not authorized for use in Army SWLAN.

The WAPs and other equipment used in a SWLAN must be afforded additional physical security controls above those afforded most wired network connections. If the device is not stored inside an approved security container, such as an IPS container, it must be physically inventoried each day.⁶³

A WIDS is a wireless network transmitter just like a WAP. As such, the design and implementation of the WIDS portion of a SWLAN must also be reviewed by the USACTTA for potential TEMPEST issues prior to the installation and operation of the WIDS equipment. It is highly recommended the review take place during the design phase of the project and prior to the procurement of any equipment.

As shown in the preceding paragraphs, the design and implementation of a SWLAN involves much more than just selecting equipment. It involves the design and implementation of local policies and procedures to ensure all aspects of network security are addressed, so the system may be accredited and approved for use. All of which involve additional costs that must be factored into the decision between using a wired and a wireless network.

3.10 VoSIP

3.10.1 General

Telephonic communications over an IP-based network have become a quick and easy means to establish voice communications worldwide. As the requirements for SIPRNET access increase, so do the requirements for a secure Voice over Internet Protocol (VoIP) system, referred to as VoSIP. VoIP is a real-time application, meaning the IP data packets containing the voice must be delivered to the distant end in a timely fashion and in sequence. Failure to do so results in the loss of quality in the communication, making the voice

⁵⁹ DISA Wireless STIG.

⁶⁰ DISA Wireless STIG.

⁶¹ DISA Wireless STIG.

⁶² DISA Wireless STIG.

⁶³ DISA Wireless STIG.

SIPRNET Technical Implementation Criteria

September 2013

received either unrecognizable and/or unintelligible. Steps taken to ensure the Quality of Service (QoS) of the VoIP link will ensure the timely sequential delivery of the voice packets.

VoSIP is merely the use of VoIP in a classified network environment. It affords users the ability to hold classified telephone conversations over their existing classified IP network. For a VoSIP call over the SIPRNET, the conversations may be at the Secret level or below.

As in a normal analog telephone call, a VoSIP/VoIP call has two main parts: the call setup and the call conversation. The call setup is not time sensitive, such as the conversation, as it uses the Transmission Control Protocol (TCP) portion of the IP standard. The time sensitive nature of the conversation lies in its use of the User Datagram Protocol (UDP) portion of the IP standard, in which the data packets are sent out, relying on the network to get them to its destination in sequence and in a timely fashion. This reliance on UDP is what makes VoIP calls susceptible to lack of QoS. In a dynamic network, such as the Internet, the NIPRNET or the SIPRNET, where the VoIP packets are mixed in with all of the other packets, there are inevitably delays and losses resulting in a low quality or even dropped conversation.

Although VoSIP will provide secure voice communication that can be used for MC functions, it will not replace the Multilevel Secure Voice (MSV) (formerly known as Defense Red Switch Network [DRSN]). MSV, with the proper COMSEC device and key material, can be used for any classification of call. It also provides many other functions that VoSIP cannot, such as preemption. While the MSV system requires special telephone sets and switching equipment with an extensive setup of the circuits linking them, VoSIP is relatively simple to establish at its most basic level.

3.10.2 Types of Calls

3.10.2.1 IP-to-IP Call

The IP-to-IP call is the simplest VoSIP call. It is a direct, point-to-point call using IP addresses instead of telephone numbers. This is accomplished by connecting a VoIP phone to the SIPRNET at each location, assigning a local IP address to each one, and “dialing” the distant end by entering the distant ends IP address.

There is no QoS involved in this type of call, and it frequently suffers from packet loss, making some conversations difficult to conduct. The call may not be able to be established if the firewalls on the SIPRNET network are not configured to permit VoSIP traffic. This type of call should not be relied upon for critical MC conversations.

3.10.2.2 Dialed Call

DISA has developed a VoSIP capability as part of the MSV for those users that do not require the full capabilities of the MSV, but do require IP-based voice communication. This system provides a global directory service that enables users to dial a telephone number to reach a specific VoSIP phone. In addition, DISA provides links between the VoSIP part of the system and the rest of the MSV. The DISA VoSIP implementation also provides the ability for a VoSIP phone on SIPRNET to communicate with a secure phone, such as a STE connected to the Public Switched Telephone Network (PSTN), via the MSV. At this time,

the system only provides QoS at the local enclave (i.e., on post). Although the global WAN portion is highly reliable, full QoS has not yet been implemented.

Similar to regular analog telephone service, DISA provides the following three classes of service for VoIP/VoSIP,⁶⁴ and Class A service requires proper justification:

- Class A: Worldwide IP-to-IP access plus IP-to-MSV access
- Class B: Worldwide IP-to-IP access
- Class C: Local enclave IP-to-IP (i.e., on post) access

The call director equipment in the DISA system translates the telephone numbers to an IP addresses similar to the way Domain Name Services (DNS) translates between the web page Uniform Resource Locator (URL) and the server IP address when browsing the Internet. The telephone numbers used must be taken from the global numbering plan used by DISA.⁶⁵

Under the global numbering plan, DISA assigns the area code and the Network Numbering Exchange (NNX) part of the 10-digit telephone number (i.e., 304-NNX-XXXX). The local VoSIP enclave administrator (typically the NEC) assigns the last four digits. The dialing format within an area code will be 7-digit, with 10-digit dialing between area codes and when dialing from VoSIP-to-MSV. Dialing from MSV-to-VoSIP will use a 12-digit dialing format, the same 10-digit format as for other calls, but prefixed with an “80” redirect number code. Table 1 shows the area codes used in the DISA numbering plan.

Table 1. VoSIP Area Codes

Region	Strategic	Tactical	Special users
CONUS	302	702 706 – SCAMPI	201
CENT	308	708 – Iraq 718 – Afghanistan	
EUR	304 301 – AFRICOM	704	
PAC	305		

AFRICOM=U.S. Africa Command; CENT=Central Command; EUR=Europe; PAC=Pacific

Likewise, the IP addresses used for the VoSIP equipment must be taken from an IP subnet issued by DISA specifically for VoSIP use⁶⁶. Use of local IP addresses on the global VoSIP network is not authorized, as they will not provide the desired connectivity due to network routing issues.

The equipment used within the local VoSIP enclave must be listed on the DISA Approved Products List (APL).⁶⁷ The current APL can be found at <https://aplots.disa.mil/>.

⁶⁴ DISA VoSIP Connection Guide.

⁶⁵ DISA VoSIP.

⁶⁶ DISA VoSIP.

⁶⁷ DISA VoSIP.

SIPRNET Technical Implementation Criteria

September 2013

To implement QoS on the local VoSIP enclave, a separate VLAN must be created for all VoSIP traffic and equipment. This VoSIP enclave must be afforded the proper network security between the enclave and where it connects to the WAN (in this case, the SIPRNET).⁶⁸ At a minimum, the network security must include an external and internal NIDS, a router with Access Control Lists (ACLs), and a firewall.⁶⁹ This network security equipment may be dedicated to the VoSIP enclave, or it may be part of a larger enclave's security perimeter, such as a SIPRNET POP provided by the post NEC.

3.10.3 Future of VoSIP

The Army Chief Information Officer (CIO)/G6 has formed a working group to make recommendations on the future of VoSIP implementations within the Army. The recommendations will be to create an Enterprise model for VoSIP with centralized management, call processing, and directory services, linked into the DISA VoSIP architecture. Pending the implementation of the Enterprise model, the Army CIO/G6 will be limiting local implementations of VoSIP within CONUS.

3.11 VTC

3.11.1 General

VTC in a classified environment may be accomplished as either a direct, one-to-one link between two suites of VTC equipment or as a bridged link using a gateway bridge device to connect two or more suites of VTC equipment. Typically, the VTC connection is via an Integrated Services Digital Network (ISDN) connection or an IP network. DISA strongly encourages the use of SIPRNET for satisfying new and existing customer secure VTC requirements. The number of ISDN resources currently available is limited to existing Defense Information Systems Network (DISN) Video Services-Global (DVS-G) users. DISA NS24 (the DISN Video Services [DVS] PMO) has determined that **new ISDN subscribers will be determined on a strict, case-by-case basis** and, in most cases, will only be authorized using a "one-in, one-out" policy whereby the DVS business office will have to identify an existing site to be removed for every new subscriber proposed.

DISA, through its DVS-G network, provides worldwide VTC bridging services. The DVS-G currently is an ISDN-based system. The new generation, DISN Video Services-Global 2 (DVS-G II), will provide IP- and ISDN-based bridging services, as well as conversion between the two. The DVS-G II is currently being fielded, and some high-level users and sites have already converted to IP-based VTC systems.

Until DVS-G II is fully operational, new users will be able to register their VTC system and will use the DISA bridging services in ISDN mode only. New users will be able to register their IP-based VTC systems on DVS-G II after the fielding is complete. Point-to-point, IP-based VTCs between two users will always be possible, since they do not use the DISA bridging services. However, VTCs that do not use the DVS-G II services will not have a guaranteed QoS, so the video and audio may degrade or be disrupted at inopportune times.

⁶⁸ DISA Enclave STIG.

⁶⁹ DISA Enclave STIG.

Additional information regarding the DVS provided by DISA is available at <http://www.disa.mil/Services/Network-Services/Video/DVS-G>.

3.11.2 Types of VTC Systems

A VTC system may be used for unclassified VTCs only, for classified VTCs only, or as a multidomain system used in both realms. When switching a VTC between domains, special procedures must be followed to prevent a security compromise. Some VTC systems perform such procedures automatically with the flip of a switch, while others must be manually recabled and reconfigured in a specific sequence.

When the VTC system is used only for classified VTCs, the VTC Coder/Decoder (CODEC) may be configured with an Ethernet and a serial V.35/RS-449/RS-530 port. The Ethernet port allows secure IP-based VTCs through SIPRNET, while the serial port allows connectivity via ISDN dial-up. [Figure A-9](#) depicts a simplified view of a VTC system equipped for ISDN- and IP-based, secure-only operation. The IP path must be disconnected when a secure ISDN VTC occurs to prevent security violations. Conversely, when a secure IP VTC occurs, the ISDN path must be disconnected.

A VTC system designed only for unclassified VTCs does not include an encryption device. The Inverse Multiplexer (IMUX) serial port connects to the CODEC serial port, and the CODEC IP port connects to the unclassified IP network. Operationally, neither the ISDN nor the IP connection must be disconnected when the other is being used.

For multidomain VTC systems, steps must be taken to ensure no information from the classified side is not leaked into the unclassified side. This includes CODEC configuration information. When switching from classified to unclassified use, all configuration information is removed from the system, and the system is reconfigured for unclassified use before being connected to the unclassified network. The CODEC is disconnected from both domains during this procedure. It is incumbent upon the VTC system owner to ensure these security measures are implemented and followed, through manual and/or automatic procedures.⁷⁰ [Figure A-10](#) shows a simplified view of a VTC system configured for ISDN- and IP-based, secure or non-secure operation. Optical isolators allow only one path, either the secure or the non-secure one, to be active at any given time.

A VTC system is typically installed in a portable, roll-around cabinet with a flat panel display on top for use in conference rooms. Mobile systems for deployment are packed in special cases. VTCs used in conference rooms or auditoriums may be mounted in fixed equipment racks or cabinets. Display devices will vary based on individual user requirements. Flat panels, fixed projectors, or portable projectors are typically used to display images. When projectors are used, fixed or portable screens can be used. For deployment situations, a small display screen built into the cases can be used. Ports are typically available on them to allow connection of portable displays or projectors and screens to the system for larger venues. Projection screen sizes should be tailored to match the characteristics of the particular VTC room or area and the audience size. Although VTC systems are typically configured with built-in audio capability, a supplemental audio system

⁷⁰ DISA VTC STIG.

SIPRNET Technical Implementation Criteria

September 2013

may need to be added to provide sufficient volume and clarity of sound, when a VTC room has poor acoustics.

3.11.3 Design and Use Considerations

A secure VTC system must be used in a way that prevents the public from viewing or overhearing classified information from the VTC. Keeping the entry door(s) closed and locked (if possible) during VTC operation is the best method to ensure compliance. If windows are present in the secure conference room, the window coverings should be drawn closed to preclude outside viewing of the VTC displays. There should also be sufficient audio attenuation through the walls and doors to prevent a person standing outside the room from understanding what is being said inside the room. The outside listener may be able to discern that someone is talking but should not be able to understand what is said.

In addition, measures must be taken to ensure collateral information is not leaked to other VTC participants. Collateral information is information that is in the VTC room but is not part of the VTC presentation or discussion. Examples of visual collateral information that could be disclosed by the VTC camera are papers lying on a table in front of a VTC participant, posters on a wall in the VTC room, papers left behind from previous meetings, or a whiteboard that is within the cameras view. Examples of audio collateral information are sideline conversations between participants in the same VTC room, a telephone conversation occurring in the VTC room but out of view of the camera, or conversations/noise entering the VTC room from outside through doors, windows, or ventilation openings. Power off the VTC system to prevent the disclosure of collateral information while the VTC room is operating in the non-secure mode. If it cannot be powered down or is not powered down, then the microphone audio must be muted, and the camera must be covered or blocked.⁷¹

For an IP-based VTC, the VTC equipment must be assigned an IP address from a subnet different from normal network users, and it must be operated in a separate VLAN reserved for VTC equipment.⁷²

Wireless network connectivity should not be used to connect VTC equipment into the network. This is especially true for secure VTCs connecting to the SIPRNET. If wireless connectivity is used, it shall meet all of the criteria and policy for SWLAN and/or WLANs.⁷³ This includes the use of wireless technology for microphones, cameras, data sharing, whiteboards, speakers, displays, and control panels used as part of the VTC system.

Similar to VoSIP, IP-based VTCs use TCP for the signaling to establish and control the VTC and UDP for the actual transmission of the video and audio portion of the VTC. The firewalls used as part of the network security equipment protecting the VTC enclave and/or the local network access to the WAN may block incoming, IP-based VTCs if not properly configured. Unfortunately, the IP ports and protocols that could be used by an IP-based VTC cannot all be allowed through the firewall due to the large number of them and the fact

⁷¹ DISA VTC STIG.

⁷² DISA VTC STIG.

⁷³ DISA VTC STIG.

that many are used by other, potentially malicious applications as well. Special application-aware firewalls, designed for use with VTCs, may be used to mitigate this issue. There are also firewall traversal technologies, based on the H.460 standard, that allow incoming and outgoing VTCs through a firewall without having to open large holes in the firewall for the numerous ports and protocols required. Firewall traversal technologies typically use a border controller on the outside of the firewall to receive incoming VTC traffic and use a gatekeeper inside the firewall. The boarder controller and the gatekeeper coordinate and pass the VTC traffic between them, through the firewall, using a very limited number of specified ports and protocols. This solution can be used with almost any firewall, whether it is an application-aware firewall or not. The use of application-aware firewalls is the preferred solution.⁷⁴

3.11.4 Secure VTC to the Desktop

Secure VTC to the desktop allows users to participate in a secure VTC while sitting at their SIPRNET computer instead of having to relocate to a VTC room. The hardware requirements for a desktop VTC are typically just a camera and a microphone. The software varies in capability, from allowing the user to participate in a one-on-one VTC to allowing the user to host a point-to-multipoint VTC with full collaboration and sharing of applications. Individual user rights on the secure computer must be addressed, as well as the configuration of the network to allow the required ports, protocols, and services to pass while maintaining network security. Local security may require the use of a push-to-talk headset, in lieu of a microphone and speaker.

The complexity and variety of hardware, software, and LANs from site to site make the discussion of a secure desktop VTC beyond the scope of this document. However, the same considerations given to securing IP-based deployment systems are warranted and required.

3.12 Thin/Zero Client

The Army will implement a centrally managed, thin/zero client, end-user computing technology. Thin/Zero Client Computing will standardize the end-user computing experience and back-end management and control. Standardization will result in increased mission effectiveness through increased accessibility, streamlined systems administration, improved security and visibility of the computing environment, and improved efficiency.

The end state is a Thin/Zero Client Computing program that provides the Army's end-user computing portion of the Joint Information Environment (JIE) to meet Army business, war fighting, and unique mission requirements. The Army will virtualize its computing environments, where appropriate, to effectively and efficiently support business and operational mission requirements. Rationalization of users as a specific type, based on mission and function, is best determined once an analysis has been conducted at the prospective post/camp/station. Refer to the [Thin/Zero Client Computing Reference Architecture](#) for specific architecture, policy, and guidance on the Thin/Zero Client Computing environment.

⁷⁴ DISA VTC STIG.

SIPRNET Technical Implementation Criteria

September 2013

3.13 IA

3.13.1 General

IA consists of the methods, procedures, and measures taken to protect the information stored, processed, accessed, or transmitted by ISs regardless of the classification of the information.

As addressed in this document, IA is specific to the construction, physical security, personnel activities, and processes related to the management and control of the SIPRNET infrastructure as discussed in this document. This document in no way implies the criteria and recommended actions described herein constitute accreditation of or for the SIPRNET infrastructure. However, failure to follow these criteria and recommendations will almost certainly result in non-accreditation, as the criteria and recommendations are taken from the policies and regulations that must be followed to achieve accreditation. [Section 4.2](#) discusses the accreditation process of this document further.

3.13.2 Information Security

All personnel involved in the generation, processing, transmission, or storage of the information are required to protect it commensurate with its classification level.⁷⁵ Classified information that is not under the direct personal control and observation of an authorized person must be appropriately guarded or stored in a locked security container or area appropriate for the classification of the material.⁷⁶

For classified information, all of the transmitted information must be encrypted using NSA Type 1 COMSEC devices,⁷⁷ be totally contained inside a CAA approved at the Secret level, or be contained inside an approved PDS.

When installing a SIPRNET system, one aspect of information security often overlooked in the initial planning is the Information Assurance Vulnerability Management (IAVM) program. This is a proactive, on-going program to ensure all ISs have all of the current required patches and updates to address vulnerabilities in its Operating Systems (OSs). The IAVM program also ensures other protective measures, such as host-based antivirus and firewalls, are current. This requires that qualified Information Technology (IT) personnel be available and assigned to these duties. The IAVM program is part of the O&M costs of a SIPRNET system.

The classified information generated, processed, transmitted, or stored in an IS is not always the only classified information that must be protected. At times, information about the IS itself may be classified, especially when the information is an aggregate of systems or a large amount of information, even though the individual bits of information may be unclassified. DISA Circular 300-115-3 gives guidance in classifying data concerning the SIPRNET.

⁷⁵ AR 380-5, Chapter 1, Paragraph 1-9, Page 3.

⁷⁶ AR 380-5, Chapter 7, Paragraph 7-4.a, Page 78.

⁷⁷ AR 25-2, Chapter 6, Paragraph 6-1.a, Page 52.

3.13.3 Equipment Security

Equipment that directly stores classified information, such as computers, Compact Discs (CDs), computer tapes, etc., is relatively easy to identify as having the capability of storing or retaining NSI. In the SIPRNET arena, such information is classified Secret. All equipment or media storing information must be clearly marked Secret⁷⁸ and must be afforded the appropriate level of physical security during use and storage.

Other types of equipment that process classified information, such as printers, photocopiers, and facsimile machines, have the capability to retain all or part of the classified information they process. Any piece of equipment with this capability to retain the classified information must be afforded the same level of protection as the information itself, and again, it must be clearly marked as such. Many of these devices retain an image of the information only on certain parts of the equipment, such as the drums in a laser printer. While it is impractical to secure a large printer or other device that is not located in an open storage area, the information potentially stored on the equipment must still be protected. This may be accomplished by removing and properly storing only the parts of the equipment that could store classified information or by ensuring the potential image has been erased. The methods and procedures for doing this vary for each type and model of equipment. Actual procedures for securing these types of equipment must be coordinated with and approved by local IA personnel.

Access to the equipment and the information stored in the equipment must be controlled using layered security techniques to ensure only authorized personnel are granted access.⁷⁹ This must be done through a combination of physical and logical security measures. The measures taken to protect the classified information and equipment are part of the physical security aspect of the accreditation process.

4.0 SIPRNET CONNECTION PROCESS

4.1 DISN Connection Approval Process (CAP)

The DISN is the aggregate of the many global services provided by DISA. The DISN is comprised of the MSV, DSN, DISN Leading Edge Services, DVS (which includes VTC capability), NIPRNET, Real Time Services (RTS) (which includes VoSIP), Cross Domain Solutions (CDS), and SIPRNET. The DISN CAP applies to a connection to any of these services. The overall flow of the DISN connection process is available at <http://www.disa.mil/Services/Network-Services/DISN-Connection-Process>.

A new system or the expansion of an existing system, which requires a new connection to the SIPRNET WAN, or is of large enough scope to require re-accreditation, must follow the CAP. An existing SIPRNET connection that is approaching the expiration date of its Authority To Connect (ATC) must also follow this DISN CAP, even in the absence of any

⁷⁸ AR 380-5, Chapter 4.

⁷⁹ Access Control STIG, Paragraph 2.

SIPRNET Technical Implementation Criteria

September 2013

type of changes to the system.⁸⁰ If the project is the expansion of an existing system using an existing connection and does not require re-accreditation, the CAP does not apply.

Supplemental information on the connection process for DVS and SIPRNET connections, along with points of contact at DISA, are given in Appendices E and I of the *DISN Connection Process Guide*, respectively.

The DISN CAP specific to VoSIP is not incorporated into the overall *DISN Connection Process Guide*. As of the date of this document, the connection process guide for VoSIP remains a separate DISA document available at <http://www.disa.mil/Services/Network-Services/Voice/VoSip>.

4.2 Accreditation and ATC

4.2.1 General

Accreditation is the official management decision by the DAA to allow an IS to operate. The DAA, in so authorizing the system to operate, accepts the risks inherent to the system and its attachment to the GIG. The DAA makes its decision based upon mission need and the recommendation from the Certification Authority (CA) as to whether the IS meets a prescribed set of security requirements.⁸¹ The accreditation goal for an IS is to achieve an Authority to Operate (ATO), which allows the system to operate. For some elements, such as with SIPRNET infrastructure, receiving proper accreditation is a precursor for receiving an ATC from DISA. This ATC is DISA approval to connect a specific SIPRNET IS to the DISN SIPRNET WAN. Upon successfully obtaining an ATO and ATC, the SIPRNET IS can be connected and made fully operational to its users.

If the SIPRNET capability at a location is currently accredited, it must be re-accredited whenever it undergoes a major change. This could be caused by a large expansion of the SIPRNET access service provided on an installation or a major change in the way SIPRNET access is provided. Regardless of any changes, the SIPRNET capability must be re-accredited every 3 years.⁸²

4.2.2 PDS Accreditation

During the planning stages and prior to the implementation or modification of a PDS, a technical review of the PDS must be requested from the USACTTA. The request for a technical review of a PDS must be submitted by a Government official (not a Contractor, although the Contractor may be listed as a Technical POC (TPOC) via e-mail and must include all information required by NSTISSI 7003, Annex C.⁸³

Requests for approval of a PDS shall be forwarded to the DAA, and must include a copy of the technical review performed by the USACTTA.⁸⁴ Note the information required in the request is more than just the layout and construction of the PDS itself. The necessary

⁸⁰ *DISA Connection Process Guide*

⁸¹ AR 25-2, Chapter 5, Paragraph 5-4.a, Page 49.

⁸² AR 25-2, Chapter 5, Paragraph 5-1.e, Page 48.

⁸³ AR 380-27, Section 3, Paragraph 3-2.

⁸⁴ AR 380-27, Section 3, Paragraph 3-3.b.

information includes aspects of physical security and operational security procedures as well. The request to approve a PDS is classified at least at the Confidential level.⁸⁵

Responsibilities concerning PDS accreditation are shared between the responsible DAA and the USACTTA, with the USACTTA being a technical reviewer to the DAA. Thus, the responsible DAA will approve a PDS, with recommendations from the USACTTA, before a system can be accredited and placed into operation. The DAA has the authority to accept the risk posed by a PDS determined non-compliant by the USACTTA⁸⁶ based upon a threat versus cost analysis. The acceptance of a risk by the DAA must occur in writing and must reference the USACTTA threat assessment.

4.2.3 C&A

All ISs must be certified and accredited.⁸⁷ The process to certify and accredit an IS has been named DIACAP. The end result of the DIACAP is to obtain certification from the DAA ensuring the system is adequately secure and is authorized to operate given the conditions and environment outlined in the DIACAP package.

The DIACAP is a five-phase process that starts with the identification of the IS and its associated security requirements (IA controls), through C&A, and finally to sustainment and eventual system decommission. It may take months to go from initial system identification and description to achieving a system accreditation decision by the DAA. DIACAP is a detail requiring a team effort. For an Army installation, the NEC (as the system owner) is the organization responsible for obtaining and maintaining an accreditation that supports the Installation Campus Area Network (ICAN) and the ISs supported by the NEC that are included as part of the ICAN accreditation.⁸⁸ The USACTTA TCR and/or PDS technical review(s) are a required part of the C&A package.

Note: The accreditation approval discussed in previous paragraphs is part of the DIACAP.

The DIACAP results in a C&A package, which is ultimately submitted for approval to the Army CA. The Army CA, as appointed by the Army CIO/G6, is the Director, Cyber Security Division, CIO/G6 (recently changed from the Director, Office of Information Assurance and Compliance, NETCOM).⁸⁹ The Army CA is responsible for making recommendations to the DAA as to whether or not the IS should be accredited. The DAA then grants or denies the ATO based on those recommendations.

When full accreditation has been achieved through the DIACAP, and the IS has been approved by the DISA CAO (the Classified CAO [CCAO] for SIPRNET ISs), DISA will issue an ATC. When the ATC has been granted, the system may be connected to the DISN SIPRNET WAN and is then fully functional and ready to provide service to the end users.

⁸⁵ AR 25-2, Chapter 6, Paragraph 6-2.f, Page 53.

⁸⁶ AR 380-27, Section 3, Paragraph 3-3.c.

⁸⁷ DODI 8510.01.

⁸⁸ AR 25-2, Paragraph 2-30.c.

⁸⁹ AR 25-2, Paragraph 3-3.n.

5.0 COST MODELING AND ESTIMATING FOR FINANCIAL PLANNING

5.1 General

After the initial design of a SIPRNET expansion project, the financial resources must be properly identified and obtained before the project can be implemented. The first step in this financial process is to estimate the cost of the project.

5.2 Cost Typing

5.2.1 General

All costs may be classified either as an expense or as an investment. Each is defined in subsequent paragraphs.

A system is comprised of a number of components that are part of, and function within, the context of a whole to satisfy a documented requirement. The system unit cost is the aggregate cost of all of the equipment and items acquired as the system.

The aggregate cost of a new end item or system shall be used to determine whether it will be considered as an expense or an investment cost. If the costs are for the modification of hardware or software components of an existing system, only the additional procurement costs will be used in the determination whether the purchase is an expense or investment.⁹⁰

5.2.2 Expense

An expense is a cost incurred to operate and/or maintain the organization, such as equipment maintenance, personal services, supplies, and utilities. Expenses must be paid for using O&M, Army (OMA) funds.

Assemblies, spares, repair parts, and other items, which have a system unit cost less than the currently approved threshold for expense/investment determination, may be purchased as an expense.

Costs that are budgeted in O&M or military personnel appropriations are expenses.⁹¹

5.2.3 Investment

An investment is a cost that results in the acquisition or addition of end items. These types of costs benefit future periods and generally are of a long-term nature, such as real or personal property. Investment costs must be paid for using Other Procurement, Army (OPA) funds.

All equipment items that have a system unit cost equal to or greater than the currently approved threshold for expense/investment determination must be purchased as an investment.

Costs that are budgeted in the Procurement, Research and Development, Test and Evaluation, or Military Construction appropriations are investments.⁹²

⁹⁰ DFAS Manual 37-100-09, Appendix A, Para D.2.a and D.2.b.

⁹¹ DFAS Manual 37-100-09, Appendix A, Para B.3.

⁹² DFAS Manual 37-100-09, Appendix A, Para B.3.

5.3 Funding Requirements and Limitations

5.3.1 General

The unit cost dollar threshold for expense and investment cost is \$250,000 for general funds and \$100,000 for Army Working Capital Funds (AWCF). The threshold for accounting and capitalization is \$100,000 and is unchanged by the threshold for expense and investment costs.⁹³

The total system unit cost determines the type of funding that must be used. The validated requirement for the system may not be fragmented or acquired in a piecemeal fashion to circumvent the expense and investment criteria policy.⁹⁴

For SIPRNET installation or expansion projects, the system unit cost includes the encryption devices, switches, PDS, cabling, and the equipment security housing. Whatever is required to satisfy the validated system requirement must be counted as part of the system unit cost.

O&M funds cannot be used to purchase any item, including replacement items, where the unit cost is more than \$250,000. If the total system unit cost exceeds \$250,000, then OPA funding must be used for procurement.⁹⁵

5.3.2 IT Procurement

When determining the total IT system cost, the labor cost to install the hardware and software must be included.⁹⁶

The cost for training and maintenance are normally separately funded using O&M funds, and must be separately funded if they are priced independently in a contract. If these costs are integral to the IT system contract cost and not broken out, they are then an integral part of the IT system cost and funded as part of the total system.⁹⁷

A LAN, CAN/MAN, or WAN are considered systems, so the total cost of all of the component parts must be used to determine the aggregate cost. Each level of network architecture (LAN, CAN, MAN, and WAN) is considered a separate system, so the costs for the procurement of each are separate⁹⁸.

When considering replacement of unusable components of a LAN system, such as substandard or non-functioning user drops, only the total cost of replacement is considered in determining the type of funds that may be used.⁹⁹ As long as the LAN system maintenance is below the threshold, O&M funds may be used.

Technology refreshment is the intentional incremental insertion of newer technology to improve reliability, improve maintainability, reduce cost, and/or add minor performance

⁹³ DFAS Manual 37-100-09, Appendix A, Para 2.

⁹⁴ DFAS Manual 37-100-09, Appendix A, Para C.2.b.

⁹⁵ DFAS Manual 37-100-09, Appendix A, Para 1.d.

⁹⁶ DFAS Manual 37-100-09, Appendix A, Para D.1.c.

⁹⁷ DFAS Manual 37-100-09, Appendix A, Para D.1.d and D.1.e.

⁹⁸ DFAS Manual 37-100-09, Appendix A, Para D.3.c(3).

⁹⁹ DFAS Manual 37-100-09, Appendix A, Paragraph D.2.b.

SIPRNET Technical Implementation Criteria

September 2013

enhancements. The addition of such technology into end items or systems as part of maintenance can be funded using O&M funds. However, technology refreshment that significantly changes the performance characteristics of the end item or system is considered a modification and must be funded as an investment using OPA funds.¹⁰⁰

Software licensing costs are considered part of the overall system cost for the initial purchase, and are thus, subject to the same funding constraints. Annual licensing costs thereafter are funded using O&M funds.¹⁰¹

The funding source for a SIPRNET project may come from a variety of sources. The SIPRNET programs listed in this document ([Section 3.1](#)) fund parts, or sometimes all, of a project. Portions of some projects may be funded by several different programs. Parts of a project that are not funded by a SIPRNET program must be funded by the organization that initiates the project. [Appendix H](#) contains a generic breakdown showing which parts of a project are funded under the MCA program.

¹⁰⁰ DFAS Manual 37-100-09, Appendix A, Paragraph C.3.d.

¹⁰¹ DFAS Manual 37-100-09, Appendix A, Paragraph D.3.c(2).

APPENDIX A. FIGURES, DRAWINGS, AND DIAGRAMS

Figures A-1 through A-10 appear on the following pages with the applicable drawings and diagrams for this document.

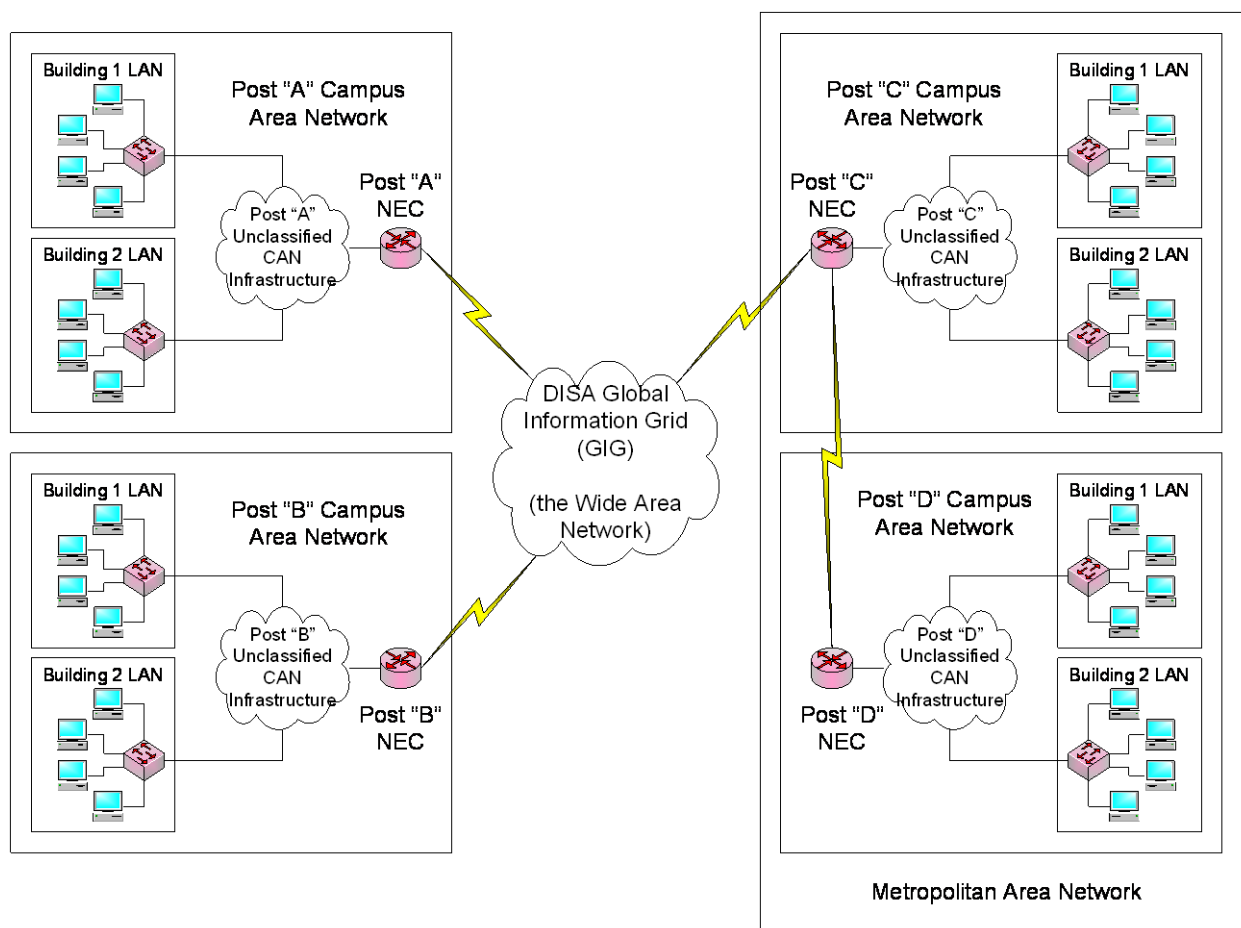


Figure A-1. Example of WAN, MAN, CAN, and LAN

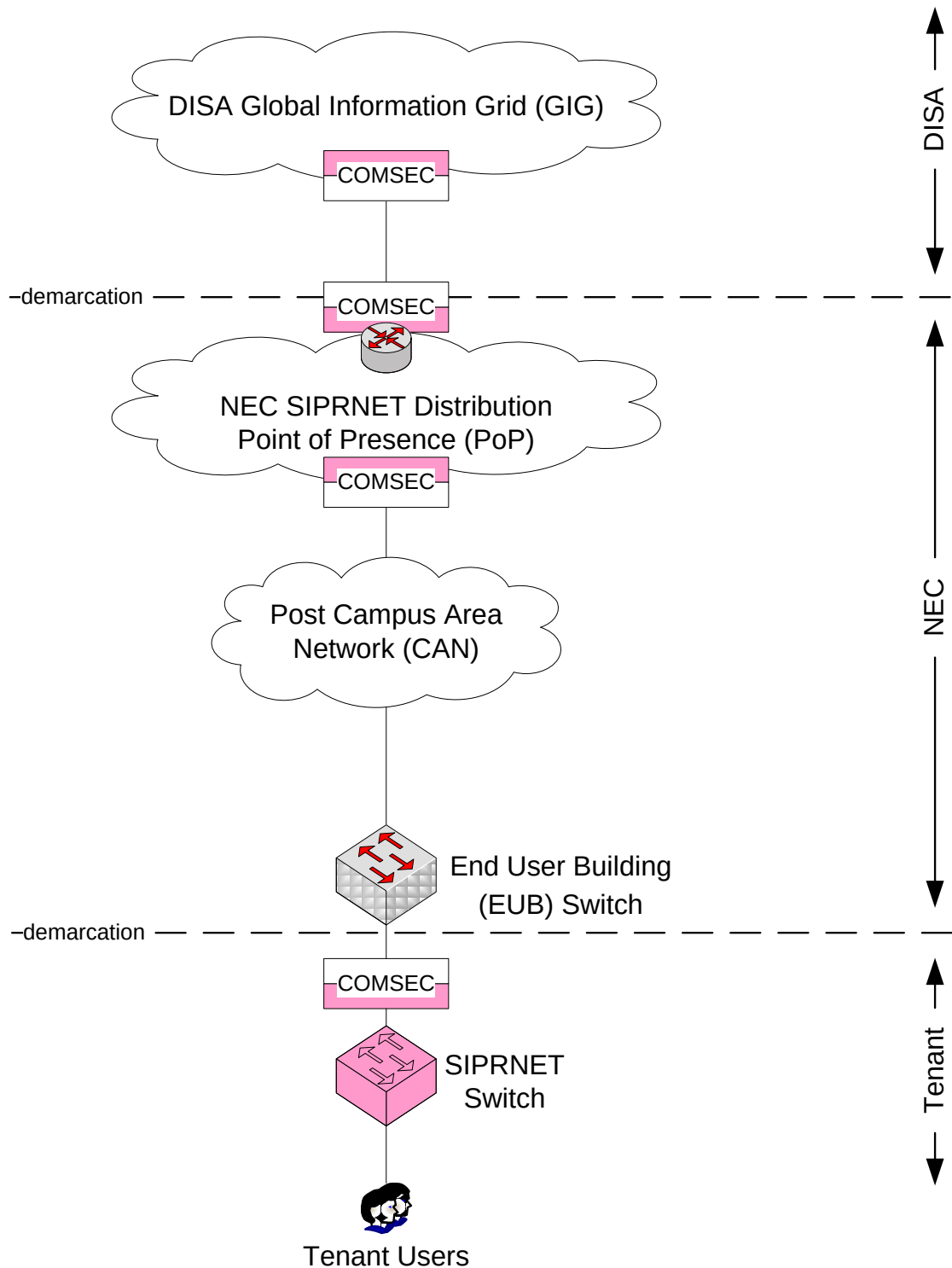


Figure A-2. SIPRNET High-Level Overview

SIPRNET Technical Implementation Criteria
September 2013

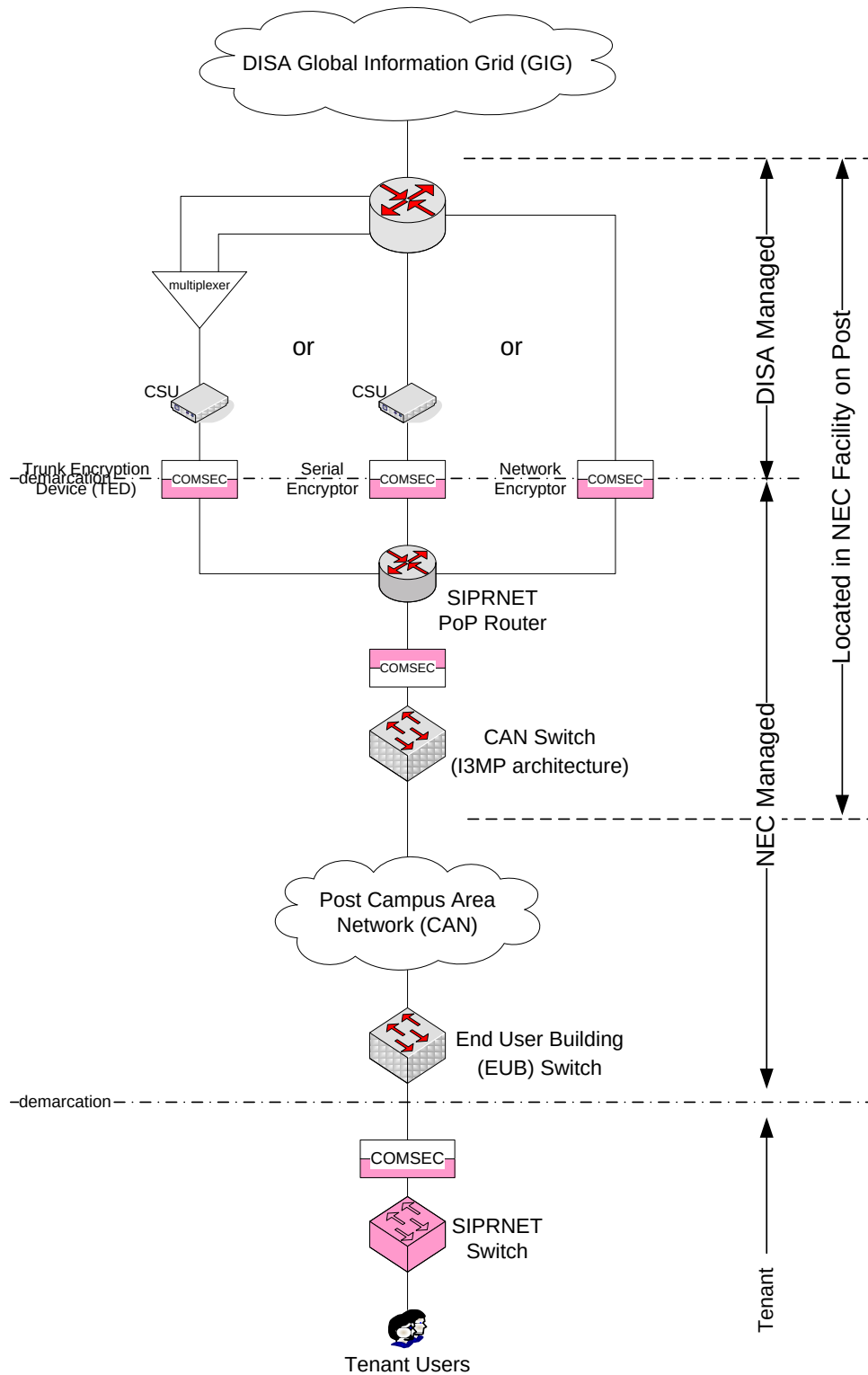


Figure A-3. DISA-Managed SIPRNET Assets

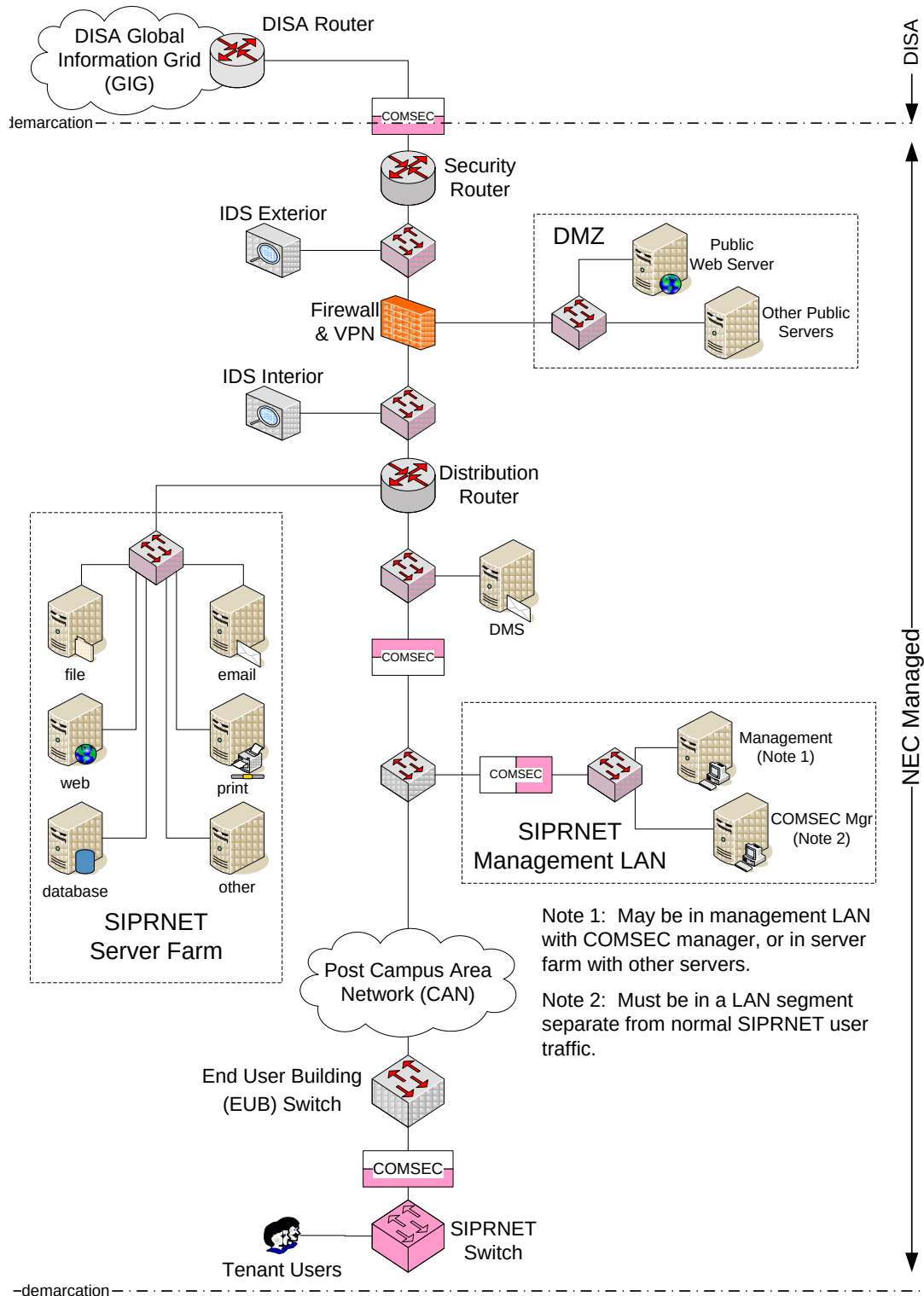


Figure A-4. NEC-Managed SIPRNET Assets

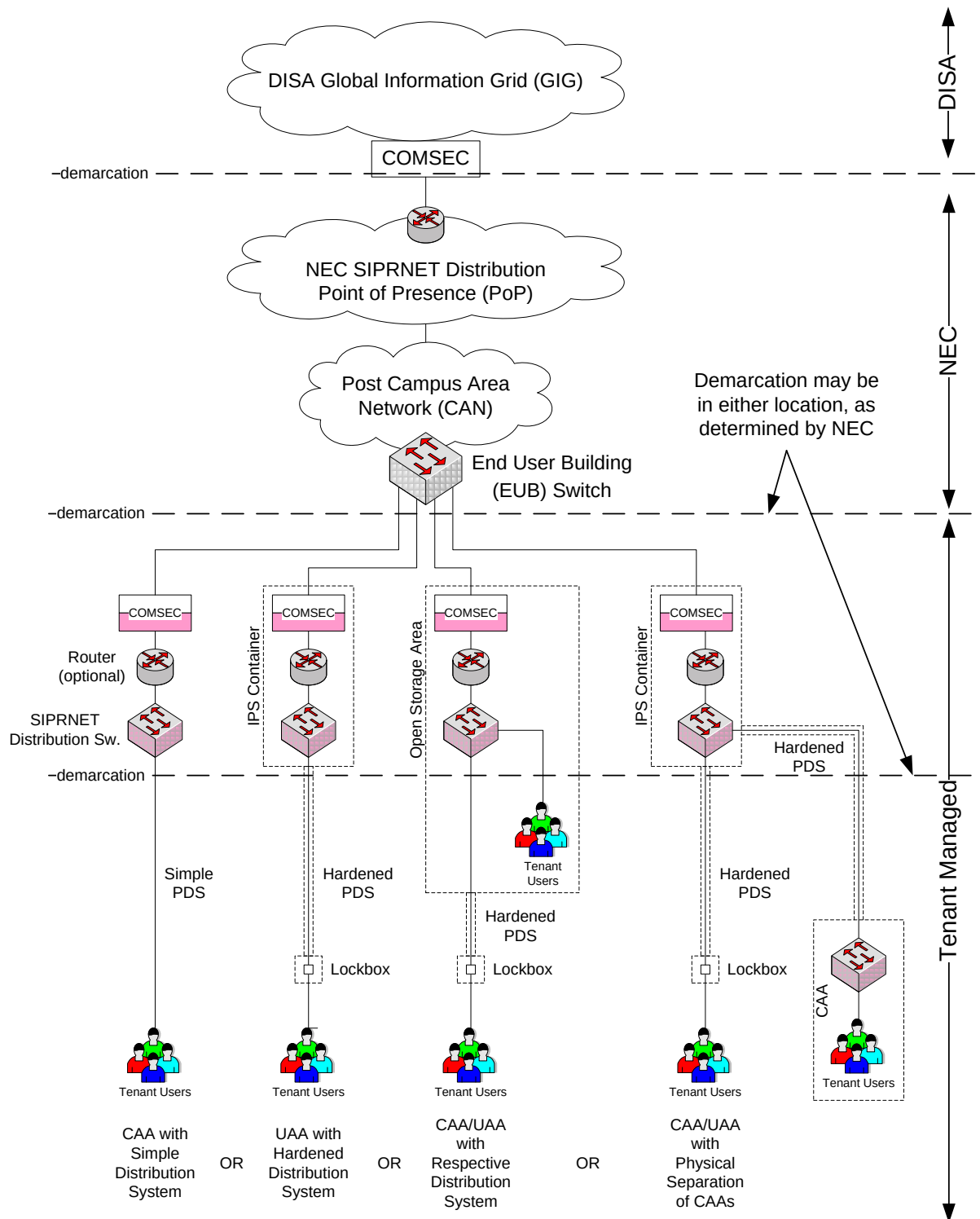


Figure A-5. Tenant-Managed SIPRNET Assets



Figure A-6. SIPRNET Dial-Up Example

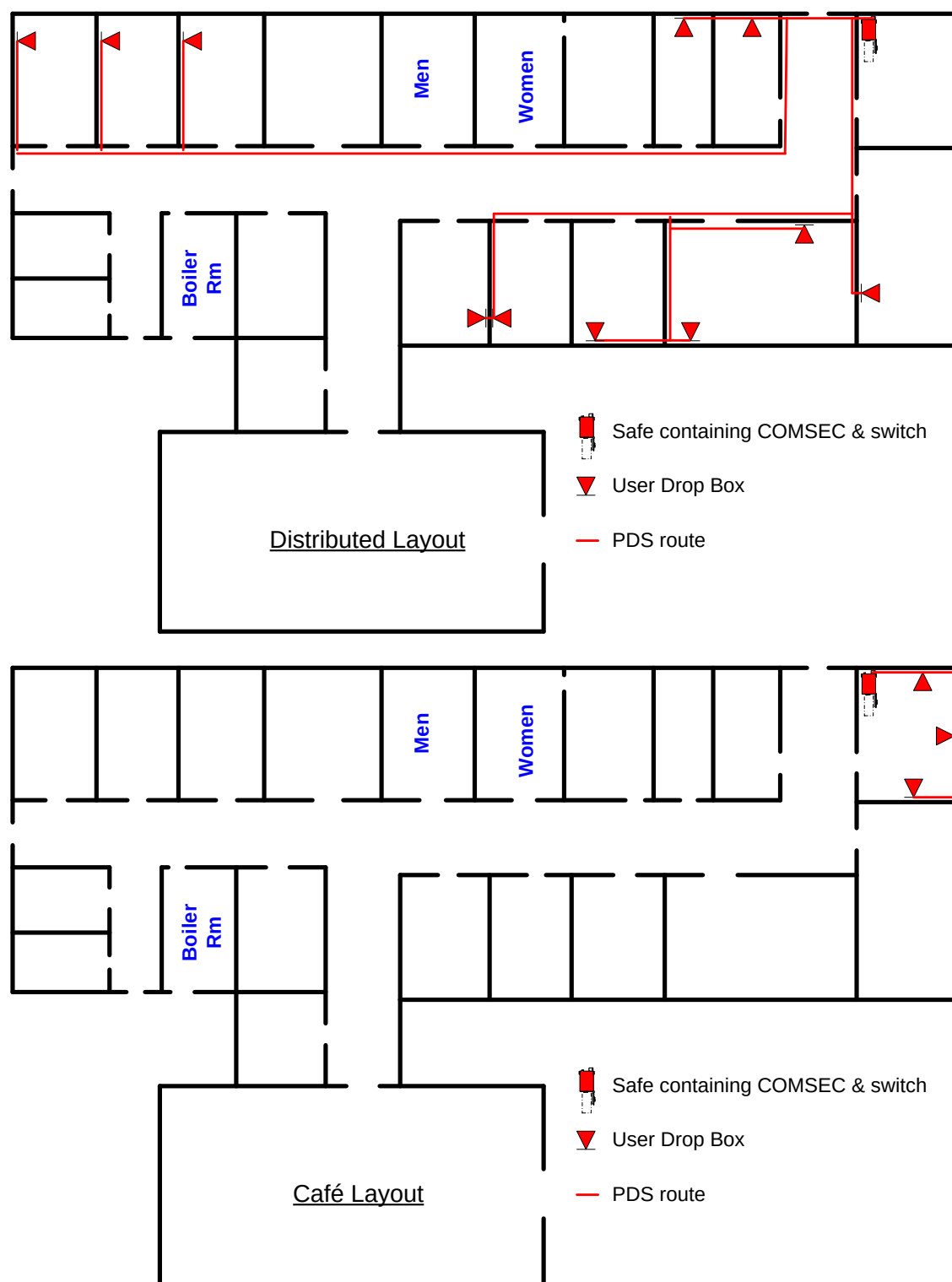


Figure A-7. PDS Physical Architecture Examples

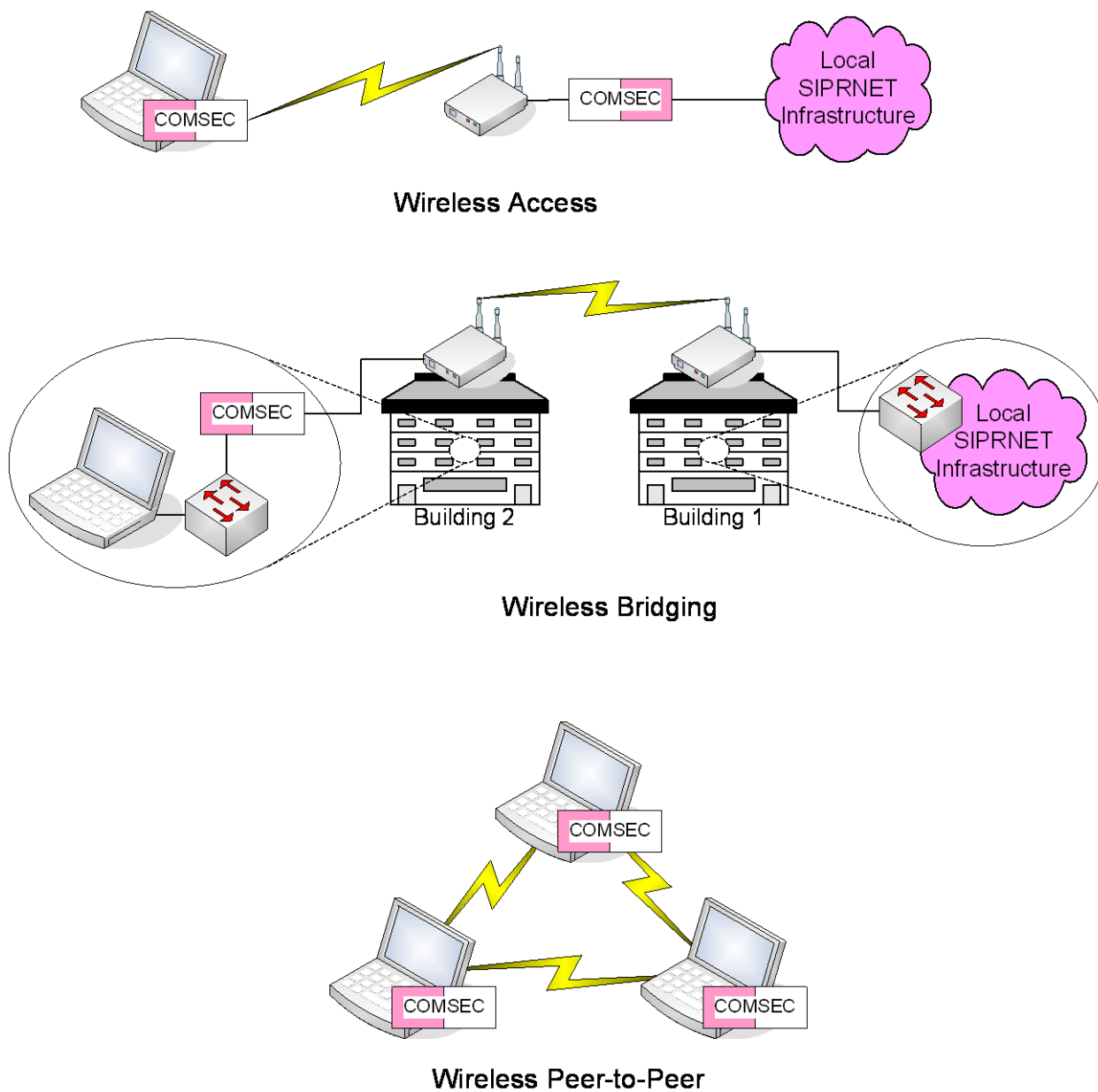


Figure A-8. Wireless Architectures

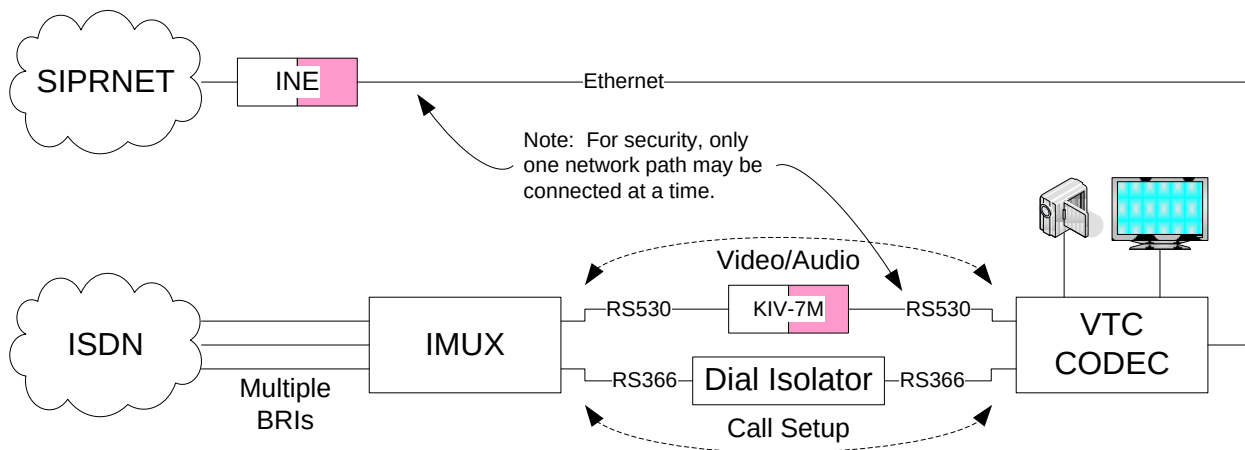


Figure A-9. Secure-Only ISDN or IP VTC

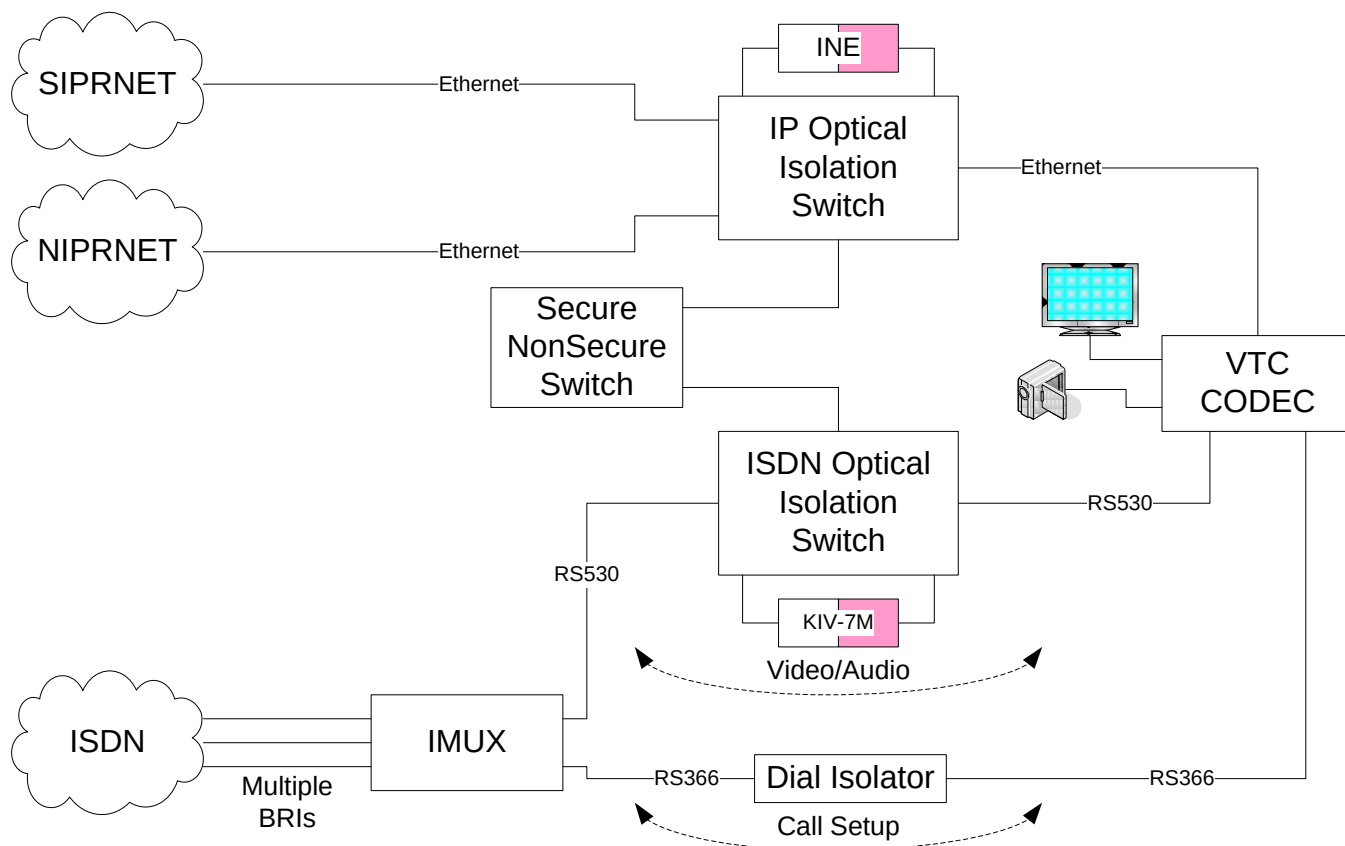


Figure A-10. Multi-Domain ISDN or IP VTC

APPENDIX B. PDS INSPECTION CHECKLIST

The PDS inspection checklist is available at <https://www.us.army.mil/suite/files/5744948>.

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APPENDIX C. SIPRNET SITE SURVEY CHECKLIST

The sample SIPRNET site survey is available at <https://www.us.army.mil/suite/files/5744948>.

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APPENDIX D. SIPRNET USER AND ALLOCATION TABLES

The SIPRNet user and allocation tables are available at <https://www.us.army.mil/suite/files/5744948>.

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APPENDIX E. SIPRNET ALLOCATIONS FOR NEW MILITARY CONSTRUCTION

The SIPRNet allocations for new military construction are available at the following web address <https://www.us.army.mil/suite/files/5744948>.

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APPENDIX F. SIPRNET PDS SPECIFICATION FOR MCA CONSTRUCTION

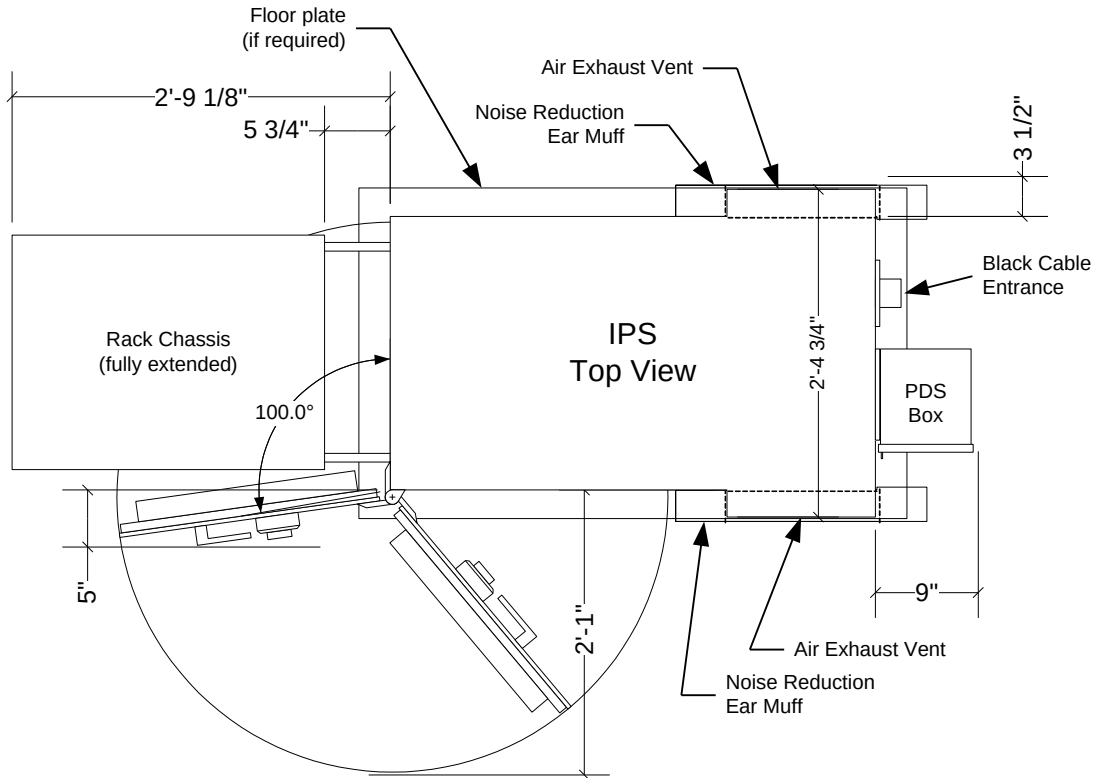
The UFGS 27 05 29.00 10, *Protective Distribution System (PDS) for SIPRNET Communication Systems*, is available at <http://www.wbdg.org/ccb/DOD/UFGS/UFGS 27 05 29.00 10.pdf>.

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APPENDIX G. SAMPLE IPS CONTAINERS WITH MOVABLE RACKS

Figures G-1 through G-4 appear on the following pages showing sample IPS containers with movable racks.

SIPRNET Technical Implementation Criteria
September 2013



NOTES

- 1 IPS shown is the Hamilton BW-2717 model.
- 2 The IPS requires certain minimum clearances around it in order to function properly, and to allow for ease of inspection. All clearance dimensions are from the surface of the IPS.
- 3 The IPS weights approximately 950 lbs empty. The four support legs on the IPS provide only 81 in² (0.5625 ft²) of floor contact area. Approximate floor loading is thus about 1,690 lbs/ft², which is above the GSA standard of 250 lbs/ft². Use of the 29"x48" floor plate provides 9.66 ft², reducing the floor loading to approximately 100 lb/ft²
- 4 The rack may be rolled out, clearing the IPS door, with the door open 100 degrees, which requires 5" of clearance from the wall.

Minimum Clearance Dimensions

Front	34"	for rack rails & frame
Hinge Side	5"	to open door for rack rollout
	25"	to fully open door
Rear	9"	for PDS box
Left Side	3 1/2"	for ear muff
Room Door	29"	to allow IPS to enter

Figure G-1. Hamilton IPS Clearance Requirements

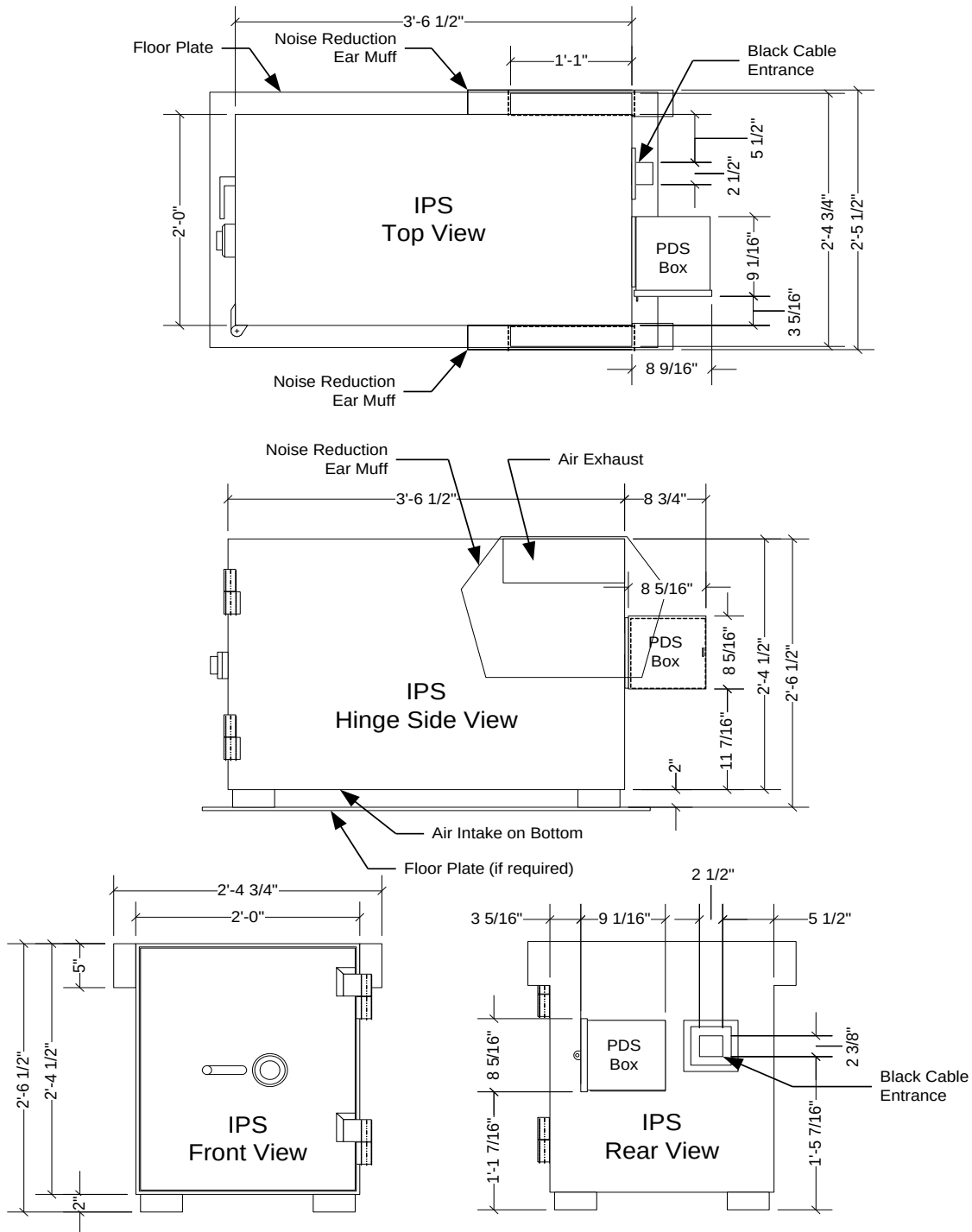
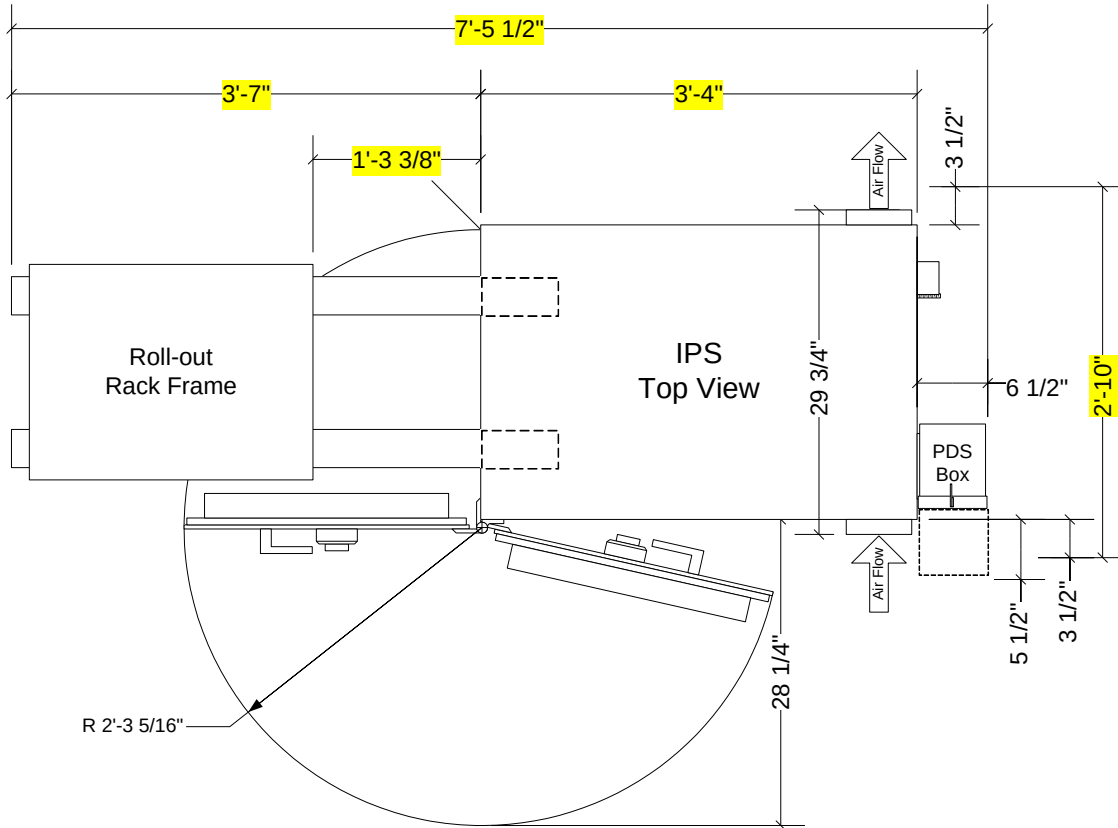


Figure G-2. Hamilton IPS Dimensions

SIPRNET Technical Implementation Criteria
September 2013



NOTES

- 1 IPS shown is the Trusted Systems TSM281WFC model.
- 2 The IPS requires certain minimum clearances around it in order to function properly, and to allow for ease of inspection. All clearance dimensions are from the surface of the IPS.
- 3 Rack may be rolled out, clearing the IPS door, with the door open 90 degrees, which requires 3 1/2" clearance. The tracks for the rack must be stored separately when not in use. The rails are 50" long and 3 1/2" wide.

Minimum Clearance Dimensions

Front	43"	for rack rails & frame
Hinge Side	3 1/2"	for air flow
	28 1/4"	to fully open door
	5 1/2"	to open PDS box door
Rear	6 1/2"	for PDS box
Left Side	3 1/2"	for air flow
Room Door	30"	to allow IPS to enter

Figure G-3. Trusted Systems IPS Clearance Requirements

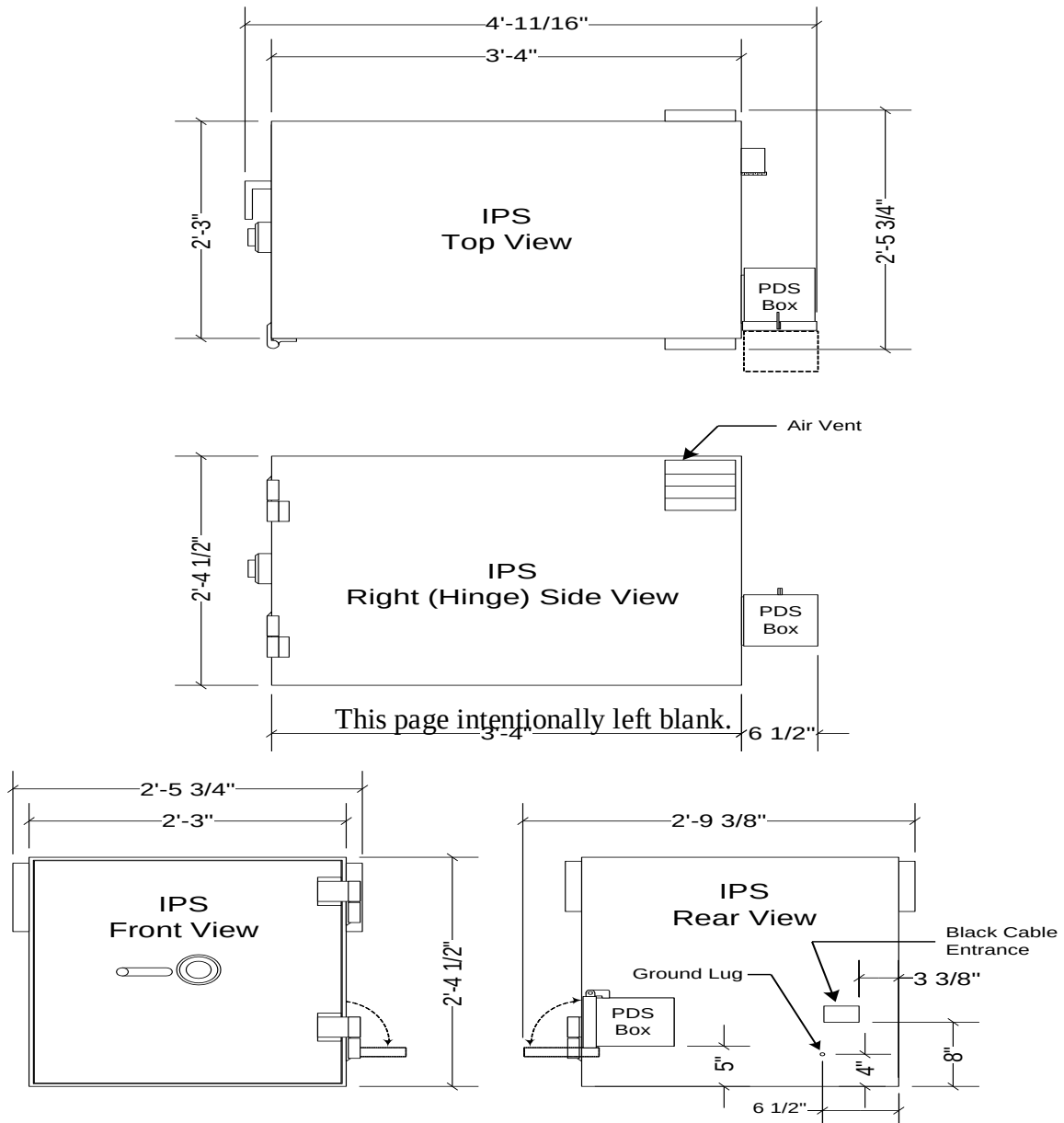


Figure G-4. Trusted Systems IPS Dimensions

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APPENDIX H. MCA INFORMATION SYSTEM COST FUNDING BREAKOUT

Figure H-1 shows the MCA Information System Cost funding breakout.

SIPRNET Information System Cost funding in MCA

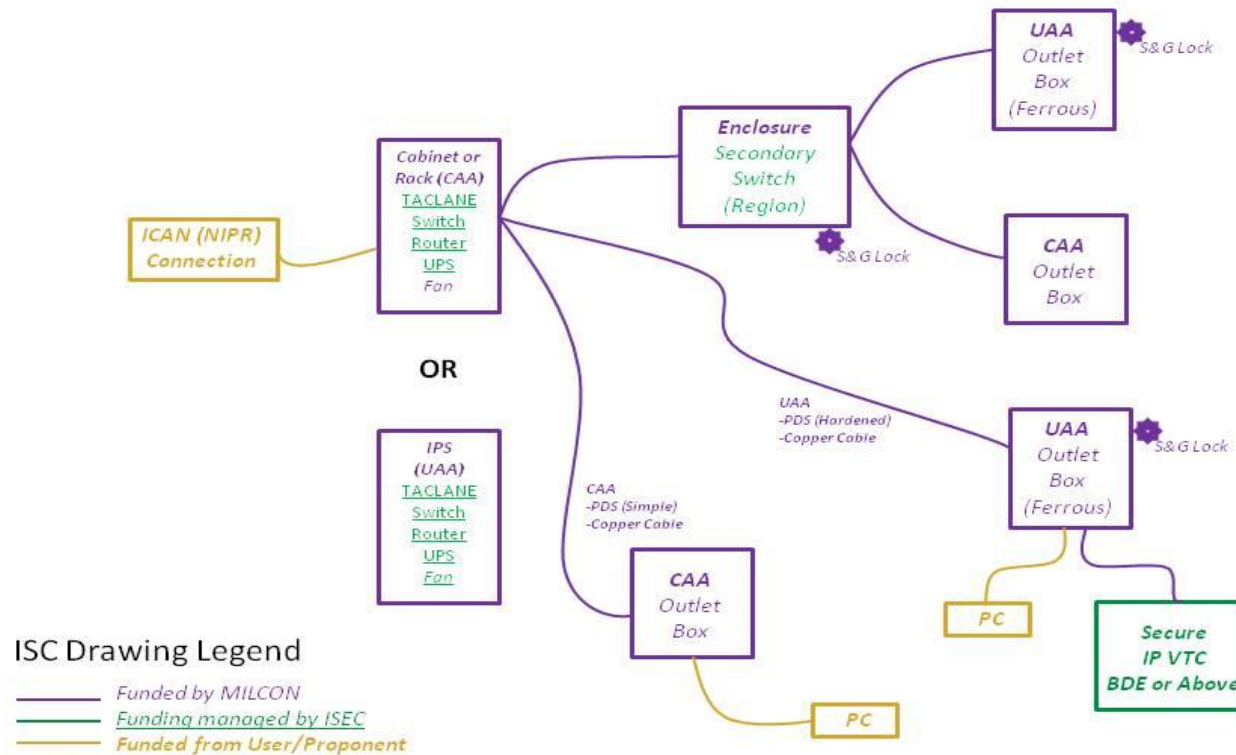


Figure H-1. MCA Information System Cost Funding Breakout

H-1

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APPENDIX I. REFERENCES

I-1.0 COMMITTEE ON NATIONAL SECURITY SYSTEMS (CNSS) PUBLICATIONS

- a. Committee on National Security Systems (CNSS) CNSS Instruction (CNSSI) 4009, *National Information Assurance (IA) Glossary*, 26 April 2010.
http://www.cnss.gov/Assets/pdf/cnssi_4009.pdf
- b. CNSS Policy 19, *National Policy Governing the Use of High Assurance Internet Protocol Encryptor (HAIPE) Products*, February 2007.
<http://www.cnss.gov/Assets/pdf/CNSSP-19.pdf>
- c. CNSS, *Index of National Security Systems Issuances*, July 2012.
<http://www.cnss.gov/Assets/pdf/CNSS-INDEX.pdf>
- d. CNSSI-4005, *Safeguarding Communications Security (COMSEC) Facilities and Materials*, 22 August 2011 (U/FOUO). (Cancels (U) NCSC Policy No.5; (U) CNSSP-10; and (U) CNSS IFOSEC 01-02, dated March 2002.
<https://csia.army.mil/Sections/COMSEC/DocNSA.aspx>
- f. IDOC-017-06, *Interim Operational Security Doctrine for the SecNet-54 Secure Wireless 802.11 a/b/g Local Area Network (SWLAN) Encryptor*, September 2006 (U/FOUO).
<https://csia.army.mil/Sections/COMSEC/DocNSA.aspx>
- g. Memorandum, SELCL-ID-P3, *SUBJECT: Interim Operational Security Doctrine for the SecNet-54 Wireless Local Area Network*, IDOC-017-06, 14 August 2007 (FOUO).

I-2.0 NATIONAL SECURITY AGENCY (NSA) PUBLICATIONS

- a. National Security Telecommunications and Information Systems Security Committee (NSTISSC) Advisory Memorandum (NSTISSAM), *TEMPEST/2-95, RED/BLACK Installation Guidelines*, 12 December 1995.
- b. NSTISSAM TEMPEST/2-95A, *Amendment to TEMPEST 2-95*, 03 February 2000.
- c. NSTISSI 7003, *Protective Distribution Systems (PDS)*, 13 December 1996 (U).
http://www.cnss.gov/Assets/pdf/nstissi_7003.pdf
- d. NSA document, *(U) Information Assurance Technical Capabilities Report*, Oct 2009

I-3.0 DOD PUBLICATIONS

- a. DOD Regulation 5200.08-R, *Physical Security Program*, 9 April 2007, with Change 1, 27 May 2009.
<http://www.dtic.mil/whs/directives/corres/pdf/520008r.pdf>
- b. DOD Directive 8100.02, *Use of Commercial Wireless Devices, Services, and Technologies in the Department of Defense (DOD) Global Information Grid (GIG)*, 23 April 2007.
<http://www.dtic.mil/whs/directives/corres/pdf/810002p.pdf>

SIPRNET Technical Implementation Criteria

September 2013

c. DOD Instruction (DODI) 8420.01, *Commercial Wireless Local-Area Network (WLAN) Devices, Systems, and Technologies*, 3 November 2009.

<http://www.dtic.mil/whs/directives/corres/pdf/842001p.pdf>

d. DODI 8510.01, *DOD Information Assurance Certification and Accreditation Process (DIACAP)*, 28 November 2007.

<http://www.dtic.mil/whs/directives/corres/pdf/851001p.pdf>

e. MIL-HDBK-1013/1A, *Design Guidelines for Physical Security of Facilities*, 15 December 1993.

http://www.wbdg.org/ccb/NAVFAC/DMMHNAV/1013_1a.pdf

f. Military Standard (MIL-STD) 188-114, Revision A, *Electrical Characteristics of Digital Interface Circuits*, September 1985 (with Notice 1, December 1991).

http://www.everyspec.com/MIL-STD/MIL-STD-0100-0299/MIL-STD-188-114A_21120/

g. *DOD Unified Capabilities Framework 2013*, Section 7.8.4, Power Backup.

<http://www.disa.mil/Services/Network-Services/UCCO/Policies-and-Procedures>

I-4.0 DEFENSE INFORMATION SYSTEMS AGENCY (DISA) PUBLICATIONS

a. *Connection Process Guide*, <http://www.disa.mil/connect>.

b. Circular 300-115-3, *Defense Information System Network (DISN) Secret Internet Protocol Routing Network (SIPRNet) Security Classification Guide*, 29 October 2009 (U/FOUO).

<https://powhatan.iie.disa.mil/disadocs/dc3001153.pdf>

I-5.0 DISA STIG PUBLICATIONS ([HTTP://IASE.DISA.MIL/STIGS](http://iase.disa.mil/stigs))

a. DISA, *Access Control in Support of Information Systems, Version 2, Release 3*, 29 October 2010.

b. DISA, *DOD Telecommunications and Defense Switched Network Security Technical Implementation Guide, Version 2, Release 3*, 30 April 2006.

c. DISA, *Enclave Security Technical Implementation Guide, Version 4, Release 3*, 28 January 2011.

d. DISA, *Network Infrastructure Security Technical Implementation Guide, Version 8 Release 14*, 26 April 2013.

e. DISA, *Voice and Video over Internet Protocol (VVoIP), Version 3, Release 2*, 27 August 2010.

f. DISA, *Video TeleConference Security Technical Implementation Guide, Version 1, Release 1*, 8 January 2008.

g. DISA, *Wireless Security Technical Implementation Guide, Version 6, Release 6*, 5 April 2013.

I-6.0 DEPARTMENT OF THE ARMY (DA) PUBLICATIONS ([HTTP://WWW.APD.ARMY.MIL](http://www.apd.army.mil))

a. AR 25-1, *Army Knowledge Management and Information Technology*, 4 December 2008.

b. AR 25-2, *Information Assurance*, 24 October 2007, Rapid Action Review, 23 March 2009.

- c. AR 190-13, *The Army Physical Security Program*, 25 February 2011.
- d. AR 190-16, *Military Police Physical Security*, 31 May 1991.
- e. AR 380-5, *Department of the Army Information Security Program*, 29 September 2000.
- f. AR 380-27, *Control of Compromising Emanations*, 19 May 2010.
- g. AR 420-1, *Army Facilities Management, Chapter 4, Army Military Construction and Non-Appropriated-Funded Construction Program Development and Execution*, 12 February 2008. Rapid Action Review, 28 March 2009.
- h. DA Pam 415-28, *Guide to Army Real Property Category Codes*, 11 April 2006.
Defense Finance and Accounting Service (DFAS) Manual 37-100-10, The Army Management Structure Fiscal Year 2012, April 2012.

<http://asafm.army.mil/offices/BU/Dfas37100.aspx?OfficeCode=1200>

- i. Technical Bulletin (TB) 380-41, *Procedures for Safeguarding, Accounting, and Supply Control of COMSEC Material*, 9 March 2006.

Note: This document is FOUO and not available online.

- j. Memorandum, DAMI-CD, *SUBJECT: Army Responsibilities for Protected Distribution Systems*, 16 August 2007.

- k. U.S. Army CIO, *Thin/Zero Client Computing Reference Architecture v1.0*, 14 March 2013.

http://ciog6.army.mil/LinkClick.aspx?fileticket=zgU_yhhjwak%3d&tabid=146

I-7.0 MISCELLANEOUS PUBLICATIONS

- a. USAISEC, Fort Detrick Engineering Directorate (FDED), *Technical Criteria for the Installation Information Infrastructure Architecture*, February 2010.
- b. USAISEC, Technology Integration Center (TIC), TR No. ELIE-ISE-TI 13-025, *Design Guide for the Implementation of Wireless Technologies*, February 2013.
- c. NETCOM Regulation 380-5, *Information Security*, 3 December 2012
- d. NETCOM 09-EC-M-0010, Best Business Practices (BBP), *Wireless Security Standards, Version 3.0*, 2 January 2009.
- e. American National Standards Institute(ANSI)/TIA 607-A-2002, *Standard Commercial Building Grounding (Earthing) and Bonding Requirements For Telecommunications*, 22 October 2002.
- f. Electronic Industries Alliance (EIA)-530, *High Speed 25 Position Interface for Data Terminal Equipment and Data Circuit-Terminating Equipment, Including Alternative 26-Position Connector*, May 26, 1992.
- g. EIA-644, *Electrical Characteristics of Low Voltage Differential Signaling (LVDS) Interface Circuits*, 1 February 2001.
- h. NFPA 70, *National Electrical Code*, 2011.
- i. Memorandum, Headquarters (HQ), U.S. Air Force/Judge Advocate, Subject: *Medical Use of Encrypted Phone Systems and SIPRNET*, 14 April 2004.
- j. Department of Defense Lock Program, <https://portal.navfac.navy.mil/go/locks>.

SIPRNET Technical Implementation Criteria
September 2013

- k. GSA Federal Specification FF-P-110J, *Padlock, Changeable Combination (Resistant to Opening by Manipulation and Surreptitious Attack)*, 11 February 1997.
- l. GSA Federal Specification FF-P-110J Amendment 1, *Padlock, Changeable Combination (Resistant to Opening by Manipulation and Surreptitious Attack)*, 20 January 2004
- m. E-mail, GSA, Mr. Pollock, *SUBJECT: QPL for Combination Padlocks and e3 class*, 4 May 2010.
- n. GSA Federal Specification FF-L-2740A, *Locks, Combination*, 12 January 1997.
http://www.gsa.gov/graphics/fas/Ffl2740a_R2FIBX_0Z5RDZ-i34K-pR.pdf
- o. GSA Federal Specification FF-L-2740A, *Locks, Combination*, 12 January 1997; Amendment 1, 25 May 2001.
http://www.gsa.gov/graphics/fas/ff-l-2740amendment1_R2FIBX_0Z5RDZ-i34K-pR.pdf
- p. GSA Federal Qualified Products List, *Products Qualified Under Federal Specification FF-L-2740A, Locks, Combination*, 10 July 2002.
http://www.gsa.gov/graphics/fas/QPL-FF-L-2740-8_R2FIBX_0Z5RDZ-i34K-pR.pdf
- q. CSLA paper, *COMSEC 101, A Survival Guide for Communications Security*, September 2009.
<https://csla.army.mil> (Requires AKO login to CSLA website.)
- r. Draft Memorandum, USACTTA, *SUBJECT: Updated Installation Guidelines for SECNET 11 Local Area Networks in U.S. Army Fixed Facilities and Systems (MMN 20073381)*.
- s. DFAS Manual 37-100-09, *SUBJECT: The Army Management Structure, Appendix A, Expense/Investment Criteria*, August 2008.

GLOSSARY. ABBREVIATIONS, ACRONYMS, AND DEFINITIONS

AC	Alternating Current
ACL	Access Control List
AES	Advanced Encryption Standard
AFRICOM	U.S. Africa Command
AKO	Army Knowledge Online
amp	Ampere
ANSI	American National Standards Institute
APL	Approved Products List
APU	Alarm Processing Unit
AR	Army Regulation
ASLAN	Assured Services Local Area Network
ATC	Authority To Connect
ATO	Authority to Operate
AWCF	Army Working Capital Funds
BBP	Best Business Practices
BICSI	Building Industry Consulting Services International
BOC	Brigade Operations Center
BTU	British Thermal Unit
C&A	Certification and Accreditation
C2	Command and Control
CA	Certification Authority
CAA	Controlled Access Area
CAN	Campus Area Network
CAO	Connection Approval Office
CAP	Connection Approval Process
CAT	Category
CCAO	Classified CAO
CCI	Controlled Cryptographic Item
CD	Compact Disc
CDS	Cross Domain Solutions
CENT	Central Command
CIK	Crypto Ignition Key
CIO	Chief Information Officer
cm	Centimeter
CNSS	Committee on National Security Systems
CNSSI	CNSS Instruction
CODEC	Coder/Decoder
COF	Company Operating Facility
COMSEC	Communications Security
CONUS	Continental United States
CSLA	Communications Security Logistics Activity
DA	Department of the Army

SIPRNET Technical Implementation Criteria
September 2013

dB	Decibel
DAA	Designated Approving Authority
DFAS	Defense Finance and Accounting Service
DIACAP	DOD Information Assurance Certification and Accreditation Process
DISA	Defense Information Systems Agency
DISN	Defense Information Systems Network
DNS	Domain Name Services
DOC	Division Operations Center
DODI	DOD Instruction
DOD	Department of Defense
DPW	Directorate of Public Works
DRSN	Defense Red Switch Network
DS3	Digital Signal Level 3
DVS	DISN Video Services
DVS-G	DISN Video Services-Global
DVS-G II	DISN Video Services-Global 2
EDP	Engineering Design Plan
EIA	Electronic Industries Alliance
EMT	Electrical Metallic Tubing
EOD	Explosive Ordnance Detachment
EUB	End User Building
EUR	Europe
F	Fahrenheit
FDED	Fort Detrick Engineering Directorate
GIG	Global Information Grid
GSA	General Services Administration
GWB	Gypsum Wallboard
HAIPE	High Assurance Internet Protocol Encryptor
HDD	Hard Disk Drive
HQ	Headquarters
HVAC	Heating, Ventilation, and Air Conditioning
IA	Information Assurance
IAM	Information Assurance Manager
IAVA	IA Vulnerability Alert
IAVM	Information Assurance Vulnerability Management
IAW	in accordance with
ICAN	Installation Campus Area Network
ICIDS	Integrated Commercial Intrusion Detection System
IDS	Intrusion Detection System
IEEE	Institute of Electrical and Electronics Engineers
IME	Individual Mobile Encryptor

SIPRNET Technical Implementation Criteria
September 2013

IMO	Information Management Officer
IMUX	Inverse Multiplexer
INE	Inline Network Encryptor
IP	Internet Protocol
IPS	Information Processing System
IS	Information System
ISDN	Integrated Services Digital Network
ISSP	Information Systems Security Program
IT	Information Technology
JIE	Joint Information Environment
JOC	Joint Operations Center
km	Kilometer
LAN	Local Area Network
lb	Pound
lbs/sq ft	Pounds Per Square Foot
LCA	Limited Controlled Area
LOAC	Law of Armed Conflict
LOM	List of Materials
LVDS	Low Voltage Differential Signaling
lux	Lumens
m	Meter
MAN	Metropolitan Area Network
MC	Mission Command
MCA	Military Construction Army
MCAR	Military Construction Army Reserve
MI	Military Intelligence
MIL-HDBK	Military Handbook
MIL-STD	Military Standard
mm	Millimeter
MPLS	Multiprotocol Label Switching
MSV	Multilevel Secure Voice
MTOE	Modified Table of Organization and Equipment
NEC	Network Enterprise Center
NETCOM	U.S. Army Network Enterprise Technology Command
NFPA	National Fire Protection Association
NIDS	Network Intrusion Detection System
NIPRNET	Unclassified but Sensitive Internet Protocol Router Network
NIPS	Network Intrusion Prevention System
NNX	Network Numbering Exchange
NOC	Network Operations Center
NSA	National Security Agency

SIPRNET Technical Implementation Criteria
September 2013

NSI	National Security Information
NSTISSAM	National Security Telecommunications and Information Systems Security Advisory Memorandum
NSTISSC	National Security Telecommunications and Information Systems Security Committee
NSTISSI	National Security Telecommunications and Information Systems Security Instruction
O&M	Operations and Maintenance
OCONUS	Outside the Continental United States
OMA	O&M, Army
OPA	Other Procurement, Army
OS	Operating System
OSI	Open Systems Interconnection
PAC	Pacific
PDS	Protected Distribution System
PMO	Program Management Office
POC	Point of Contact
POE	Power over Ethernet
POP	Point of Presence
PSTN	Public Switched Telephone Network
PVC	Polyvinyl Chloride
QA	Quality Assurance
QC	Quality Control
QoS	Quality of Service
RF	Radio Frequency
RTS	Real Time Services
RU	Rack Unit
SCIF	Sensitive Compartmented Information Facility
SDD	Secure Data Device
SIPRNET	Secret Internet Protocol Router Network
SMS	Security Management Server
SNMP	Simple Network Management Protocol
SOF	Special Operations Forces
SONET	Synchronous Optical Networking
SRM	Sustainment, Restoration, and Modernization
STC	Sound Transmission Category
STE	Secure Terminal Equipment
STIG	Security Technical Implementation Guide
SWLAN	Secure Wireless Local Area Network
T1	Transmission Carrier 1

TACLANE	Tactical Local Area Network Encryptor
TB	Technical Bulletin
TCP	Transmission Control Protocol
TCR	TEMPEST Countermeasure Review
TDA	Table of Distribution and Allowances
TEMF	Tactical Equipment Maintenance Facility
TGB	Telecommunication Grounding Busbar
TIA	Telecommunications Industry Association
TIC	Technology Integration Center
TMGB	Telecommunications Main Ground Busbar
TPM	TEMPEST Program Manager
TPOC	Technical POC
TR	Telecommunications Room
UAA	Uncontrolled Access Area
UDB	User Drop Box
UDP	User Datagram Protocol
UPS	Uninterruptible Power Supply
URL	Uniform Resource Locator
USACE	U.S. Army Corps of Engineer
USAISEC	U.S. Army Information Systems Engineering Command
USACTTA	U.S. Army Central TEMPEST Technical Authority
VAC	Volts Alternating Current
VoIP	Voice over Internet Protocol
VoSIP	Voice over Secure Internet Protocol
VPN	Virtual Private Network
VTC	Video Teleconference
VVoIP	Voice and Video over Internet Protocol
WAN	Wide Area Network
WAP	Wireless Access Point
WIDS	Wireless Intrusion Detection System
WiMAX	Worldwide Interoperability for Microwave Access
WLAN	Wireless Local Area Network
WPA2	Wireless Protected Access 2

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**ATTACHMENT 1. MEMORANDUM, U.S. AIR FORCE JUDGE
ADVOCATE, SUBJECT: MEDICAL USE OF ENCRYPTED PHONE
SYSTEMS AND SIPRNET, 14 APRIL 2004**

The U.S. Air Force Judge Advocate Memorandum follows.

SIPRNET Technical Implementation Criteria
September 2013



DEPARTMENT OF THE AIR FORCE
HEADQUARTERS UNITED STATES AIR FORCE
WASHINGTON, DC

14 APR 2004

MEMORANDUM FOR HQ AFCA/JA

FROM: HQ USAF/JA
14210 AF Pentagon
Washington D.C. 20330-1420

SUBJECT: Medical Use of Encrypted Phone Systems and SIPRNET

- References:** (a) Request for Legal Review of Medical Use of Encrypted Phone Systems and SIPRNET, 30 Sep 03
- (b) Permissibility of Encrypting Protected Medical Information in Communications to and from Fixed Medical Treatment Facilities under the Law of Armed Conflict, DoD/GC (Health Affairs), 28 Mar 03
- (c) Geneva Convention for the Amelioration of the Condition of the Wounded and Sick in the Armed Forces in the Field (12 August 1949) (GWS)
- (d) Geneva Convention for the Amelioration of the Condition of the Wounded, Sick and Shipwrecked of Armed Forces at Sea (12 August 1949) (GWS Sea)

Reference (a) requests a legal review of the use of SIPRNET and encrypted phone systems by medical facilities. This legal review discusses those provisions of the Law of Armed Conflict (LOAC) that affect secure electronic communications to and from medical facilities. We conclude that the installation and use of SIPRNET terminals and encrypted phone systems in medical facilities is permitted by LOAC, but only if the equipment is used for the exclusive purpose of medical operations.¹

Medical units and establishments organized for medical purposes, whether fixed, mobile, permanent or temporary, are protected from attack if they are exclusively engaged in the search for, collection, transportation, diagnosis or treatment, including first-aid treatment, of the wounded, sick and shipwrecked, or the prevention of disease. Medical facilities include hospitals, blood transfusion centers, preventive medical centers and institutes, medical depots, and medical and pharmaceutical stores of such units.²

¹ Medical operations are limited to those actions reasonably necessary for the medical facility to perform its mission, which under LOAC is the search for, collection, transportation, diagnosis and treatment of wounded, sick and shipwrecked persons, and the prevention of disease.

² Article 19, GWS.

If a medical facility is used for acts inconsistent with its medical mission and protected status, or otherwise engages in acts harmful to the enemy, it may lose its protected status. Loss of its protected status can occur only after due warning has been made and the warning has been disregarded.³ The fact that a medical facility is protected by pickets, sentries, or armed escorts does not deprive it of the protection guaranteed by the Geneva Conventions.⁴

Article 34 of GWS (Sea) provides that "hospital ships may not possess or use a secret code for their wireless or other means of communication." However, there is no corresponding prohibition that applies to land-based medical facilities. Therefore, the obligation of the U.S. armed forces is to maintain medical facilities for the exclusive purposes of providing health care and to avoid using those facilities for any acts harmful to the enemy. Based on the foregoing, DoD/GC determined LOAC does not prohibit encrypted Internet/NIPRNET transmission of medical information to or from land-based medical facilities.⁵

DoD/GC also noted that in order to make it clear to enemy forces that a medical facility is transmitting only medical information (and not intelligence information), there may be policy reasons to set up communications so that they flow unencrypted in a manner that would allow the medical facility to demonstrate its transmissions contain only medical information.⁶ Other alternatives would be to make provisions for a neutral third party to verify that a medical facility's transmissions contain only medical information, or to use systems that could be easily disconnected if necessary. However, these are policy considerations not legally required by LOAC.

The use of SIPRNET⁷ and encrypted phone communications⁸ by medical facilities relies on the same international law analysis as the use of encrypted Internet/NIPRNET systems. Consequently, we conclude that LOAC does not bar a medical facility from installing a SIPRNET terminal or an encrypted phone system to transmit or receive encrypted information, *so long as the systems are used only for medical purposes*. The presence of SIPRNET and encrypted phone systems in a medical facility does, however, create some unique policy considerations.

Because the SIPRNET was initially intended as a command and control tool, adversaries may be skeptical of U.S. claims that its medical facilities with SIPRNET terminals are using the

³ Article 21, GWS.

⁴ Article 22, GWS.

⁵ See *Permissibility of Encrypting Protected Medical Information in Communications to and from Field Medical Treatment Facilities under the Law of Armed Conflict*, OSD/GC (Health Affairs), 23 Mar 03. This opinion supersedes earlier opinions by AFJAG and other AF offices regarding the use of encrypted communications by medical facilities.

⁶ This may, of course, subject the data to interception by a belligerent sophisticated enough to satisfy itself that the transmissions are solely medical in nature.

⁷ The SIPRNET has matured to be the core of our warfighting command and control capability. Subscribers within the DoD and other Government Agencies are able to use the SIPRNET for passing data at the Secret Not Releasable to Foreign Nationals classification level. Although the SIPRNET was originally designed as a command and control system, it is fast becoming the de facto standard of preferred data services, even over the NIPRNET.

⁸ Encrypted phone equipment used by DoD typically consists of either the Secure Telephone Unit - Third Generation (STU-III) or the newer digital-based Secure Terminal Equipment (STE). The STU-III is a low-cost, use-friendly, secure telephone device. The terminals are designed to operate reliably, with high voice quality, as both ordinary telephones and secure instruments over the existing public switch telephone network. The Secure Terminal Equipment is the evolutionary successor to the STU-III. The STE program improves communications by changing out the analog STU-III products with digital-based STE products.

terminals only for medical purposes. Therefore, for policy reasons commanders should consider some of the same safeguards suggested with respect to general encrypted communications, such as making provisions for a neutral third party to verify that the system is being used only for medical purposes, or using systems that can be easily disconnected if necessary. It may also be useful to request State Department assistance in communicating to opposing belligerent governments the legitimate purposes of having secure communications in medical facilities.⁹ Commanders should also consider the practical security issues associated with placing classified equipment and information in medical facilities, such as the necessity for a secure repository and a secure area with limited access. Medical facilities that are in areas where they risk being captured will have to take reasonable measures, within the bounds of LOAC, to protect the encryption equipment and classified information.¹⁰

Use of encrypted phone systems and SIPRNET by medical facilities does not violate LOAC as long as information transiting the system is in furtherance of the medical mission. As with encrypted Internet/SIPRNET transmission of medical information, there are likely many ways in which the risks associated with using SIPRNET and encrypted phone equipment in medical facilities can be addressed, mitigated, and balanced against the potential benefits. We defer on such policy considerations, and leave it to MAJCOM Communications and Information Directorates to appropriately consider these issues when deciding whether the installation of specific encrypted equipment in a particular medical facility is appropriate. We do recommend that a thorough legal and policy review be conducted in each case before a system is installed. Reviewing authorities may want to require medical facilities to submit a detailed justification that includes information such as why the system is required, why removable equipment will not suffice, the particular medical uses for which the system will be used, a description of the type of medical information that will be encrypted, and the steps that will be taken to ensure the equipment will be used only for medical purposes. It may also be useful for general guidelines to be published by the appropriate agencies regarding these issues.


THOMAS J. FISCUS
Major General, USAF
The Judge Advocate General

cc:
MAJCOM/IAS
AFJAGS
SAF/GC
HQ AF/SG

⁹ For example, many aspects of military medical operations, such as casualty reporting, readiness status reports, and the protection of certain patient information under U.S. law may require medical facilities to use secure communication channels.
¹⁰ The mere presence of encryption equipment in a medical facility creates some risk because an enemy may be skeptical of U.S. assurances that the equipment is being used solely for medical purposes. Likewise, if medical personnel destroy or disable the equipment to prevent it from falling into enemy hands, the enemy might question whether such actions fall within the medical personnel's humanitarian duties—putting their protected status at risk. However, LOAC is ambiguous on this issue. There is no clear legal authority that such action is prohibited, and it is arguably more consistent with the principles of LOAC to prevent equipment used solely for medical purposes from being converted to broader military intelligence uses when captured by an adversary.